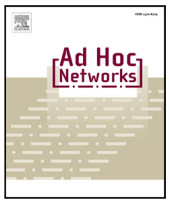
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Survey paper



A complete survey on artificial intelligence based resource allocation for sixth generation mobile

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ABSTRACT

Sixth generation (6G) mobile networks promise unprecedented connectivity with ultra-high data rates, near-zero latency, and high reliability for applications like autonomous systems and extended reality. However, managing diverse resources, communication, computing, and caching (3C), poses significant challenges. Artificial intelligence (AI) is the key to automating and optimizing resource management in 6G. This review compares traditional model-based methods with adaptive data-driven approaches, covering computing and caching resource management alongside radio resources within the edge-cloud continuum. Crucially, the paper examines big AI models (BAIM) and the shift toward agentic AI for holistic, autonomous network automation. To address the opacity of these complex models, we highlight explainable AI (XAI) and digital twins (DTs) as an essential, combined trust and validation layer. Together, they ensure transparency, mitigate the computational overhead of real-time explanations, and enable safe training for critical functions like network slicing and multi-access edge computing (MEC) computation offloading. Finally, key challenges and future research directions for AI integration in next-generation wireless networks are outlined.

1. Introduction

The evolution of mobile telecommunications began with first generation (1G) analog systems in the 1980s, progressing through digital communication and high-speed data in later generations. The introduction of the Long Term Evolution (LTE) standard in 2009 marked fourth generation (4G) networks, offering enhanced broadband services. Today, wireless networks are shifting from 4G to fifth generation (5G) systems based on the new radio (NR) standard. Driven by demand for enhanced mobile broadband, 5G incorporates higher frequency bands, private networks on unlicensed spectrum, and flexible topologies. The rise of internet of things (IoT) devices and the need for ultra-low latency and high reliability have further improved user experiences. Building on these advancements, sixth generation (6G) networks aim to meet emerging technological and societal needs by 2030 [1–5]. For clarity, a comprehensive list of acronyms used throughout this paper is provided in Table 1.

Modern society is increasingly data-driven, powered by pervasive sensors, automation, and autonomous systems. Artificial intelligence (AI) deployed across cloud, edge, and fog computing environments enables intelligent services and drives economic innovation. The forthcoming 6G networks aim to realize the internet of everything (IoE) by connecting people, devices, sensors, vehicles, and robots. AI will be integrated into the network core and edge to deliver real-time intelligence, self-management, and adaptive communication. A significant innovation in 6G is the space–air–ground integrated network (SAGIN) [6], which unifies terrestrial, aerial, and satellite systems into a single network fabric. SAGIN extends coverage to remote and underserved areas, enhances capacity, and supports advanced applications like autonomous systems and extended reality (XR). This integration addresses the complexity of future networks and enables intelligent resource management essential to 6G [6]. In this context, AI and machine learning (ML) are vital for autonomous network operation, supporting dynamic spectrum allocation, routing optimization, energy efficiency, and security. The strategic role of AI in next-generation networks is underscored by extensive ongoing global research in this field [7–15].

6G networks promise ultra-high data rates by utilizing advanced spectrum resources such as millimeter wave, terahertz (THz), and visible light bands. Managing scarce and dynamic resources is challenging due to diverse service requirements and network complexity. Traditional approaches, e.g., optimization, heuristics, and game theory, struggle to meet the scale of 6G systems. The massive connectivity, data volumes, and heterogeneous user demands further complicate resource management [16]. Thus, integrating AI and ML is essential for efficient and scalable network operation [17].

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AI is a multidisciplinary field encompassing ML, biologically inspired computing, and expert systems [18], applied widely to clustering, classification, optimization, and regression tasks. Prior studies [19–21] highlight AI's role in enhancing management intelligence in 5G, focusing on resource management, mobility, and service delivery. Looking forward, AI addresses key challenges in future wireless systems such as massive connectivity, large-scale data analytics, and diverse quality of service (QoS) and quality of experience (QoE) requirements. AI-driven allocation and multiplexing improve resource utilization, while data mining exposes traffic and mobility patterns.

In this review, we define "AI-driven" resource management through two distinct levels of autonomy. The first level, AI-assisted optimization, describes systems where AI offers analytical insights or enhances specific optimization tasks, such as channel estimation, while final decision control remains under rule-based mechanisms or human oversight. In contrast, the second level, fully AI autonomous decision-making, aligns conceptually with the emerging paradigm of agentic AI. This involves AI agents operating within continuous perception-action loops, independently perceiving network states, reasoning to achieve predefined operator goals, and utilizing network functions as tools to execute resource allocations (e.g., via advanced Deep Reinforcement Learning (DRL) or Foundation Models) without real-time human intervention [22]. Furthermore, we categorize 6G resources using the unified communication, computing, caching (3C) framework [16]. The 3C framework refers to the unified model of communication, computing, and caching resources in modern and future wireless networks like 5G and 6G. It provides a structured approach to integrate and optimize these interdependent elements for efficient service delivery, such as low-latency applications and immersive experiences. The communication domain includes spectrum resources (e.g., Radio Frequency, THz, and Optical bands), alongside power management and advanced access technologies such as cognitive radio (CR). Conversely, computing resources comprise the processing power distributed throughout the network continuum, spanning from end devices to the edge and the core cloud. Lastly, caching comprises distributed storage resources throughout the edge–cloud continuum. Network slicing is viewed not as a separate resource but as a virtualization technology that orchestrates the 3C resources to satisfy service level agreements (SLAs).

Despite its potential, AI–based resource allocation in 6G networks faces challenges from network scale, complexity, and heterogeneity. Fig. 1 outlines key challenges and AI-enabled solutions for next-generation wireless systems. The rapid growth of 6G, marked by dense connectivity, non-terrestrial architectures, and IoE ubiquity, requires adaptive, low-latency AI algorithms to manage dynamic mobility and traffic. Computationally intensive models and large datasets burden resource-limited devices, increasing latency. AI integration also raises security and privacy risks such as data poisoning and adversarial attacks, calling for defenses like blockchain and federated learning (FL). Ensuring interoperability across terrestrial, aerial, and satellite networks adds further complexity. Additionally, AI's high computational cost contrasts with the sustainability goals of 6G, underscoring the need for energy-efficient algorithms [23]. Finally, the opacity of AI decision processes raises trust and accountability concerns. Explainable AI (XAI) is therefore essential to promote transparency, interpretability, and safe deployment in 6G communication systems [24].

The evolving challenges highlight the need for a systematic review of AI-based resource allocation strategies and their integration with XAI frameworks. This paper addresses this gap by providing a comprehensive analysis of current research trends, methodologies, and open issues in AI-driven and explainable resource management for 6G networks.

1.1. 6G network architecture

The 6G network architecture typically comprises three layers: the top *intelligent service layer*, the middle *virtualized network layer*, and the bottom *heterogeneous edge–cloud infrastructure* (Fig. 2). This modular, AI-driven design replaces the monolithic 5G core, enabling terabit data rates, ultra-low latency, and massive connectivity [25].

- Top layer – Service and intelligence: Translates user and industry requirements into tasks or intents, such as XR slices or industrial control. It hosts global AI, analytics, and digital twin functions for closed-loop optimization, orchestrating network slices and policies across the 6G core and radio access network [25,26].
- Middle layer – Cloud-native network: A microservice-based core that separates control and user planes, enabling fine-grained slicing and network federation. It manages functions such as mobility, session management, security, and 6G Recursive User Plane Architecture (6G-RUPA), deployable near users or in the cloud for performance and flexibility [25,26].
- The bottom layer of the 6G architecture aggregates diverse resources across multi-tier edge–cloud computing and an integrated space–air–ground–sea (SAGS) fabric. This infrastructure extends beyond traditional boundaries by unifying satellite, aerial, terrestrial, and maritime resources into a single cohesive network. Within this ecosystem, super edge nodes (SEN) converge radio access, computing, and storage near the user to enable ultra-low latency. Crucially, to manage the complexity of this multi-domain environment, recent frameworks propose "AI-Native Task Orchestration" for space–air–ground–sea integrated networks (SAGSINs) [27]. By autonomously managing collaborative computing across terrestrial and non-terrestrial nodes, this AI-driven approach dynamically optimizes heterogeneous resources to ensure seamless service continuity across diverse environments [6,25,26].

For more technical details, please refer to [25,26].

1.2. Emerging resource types in 6G

In 6G networks, communication resources include the utilization of new spectrum bands such as THz and visible light communication, enabling ultra-high bandwidth and data rates. Technologies like intelligent reflecting surfaces (IRS) and non-orthogonal multiple access (NOMA) are employed to enhance spectral efficiency. Recent advancements in this domain include intelligent optimization schemes such as the Advanced Non-Convex Non-Smooth (ANNS) framework [28], which is specifically designed to optimize IRS-aided next-generation networks. Additionally, unmanned aerial vehicles (UAVs) are integrated as aerial communication platforms, extending network coverage and flexibility. Cell-free massive multiple-input multiple-output (MIMO) systems and full-spectrum uniform cell-free network architectures are also deployed to provide seamless coverage and improve network performance across diverse environments [29]. Furthermore, SAGIN integrates satellite, aerial, and terrestrial networks to provide seamless, ubiquitous communication coverage, especially in remote or underserved areas. It supports dynamic spectrum sharing and heterogeneous links to enhance connectivity, capacity, and reliability across space, air, and ground layers [6].

Computation resources in 6G are expected to be highly distributed and intelligent, utilizing pervasive multi-level AI across centralized data centers, edge servers, and end-user devices within the edge–cloud continuum. SEN are proposed to integrate connectivity, computing, AI, and data management, enabling near real-time resource orchestration and efficient task handling. SAGIN enables distributed computing by deploying SEN throughout space, air, and ground domains. This facilitates computation offloading from devices to nearby edge servers or aerial platforms, reducing latency and improving processing efficiency in 6G networks. Task-centric AI-driven automation is also being developed to dynamically allocate resources based on the specific needs of services and applications, enhancing the overall efficiency and responsiveness of the network [30].

Caching resources in 6G are enhanced by edge caching enabled through the edge–cloud continuum, which helps reduce latency and alleviate the load on backhaul networks. By leveraging the distributed

Table 1
A list of commonly used acronyms in this paper.

Acronym	Definition	Acronym	Definition
1G	First Generation	3C	Communication, Computing, Caching
4G	Fourth Generation	5G	Fifth Generation
6G	Sixth Generation	6G-RUPA	6G Recursive User Plane Architecture
ACO	Ant Colony Optimization	ADR	Aggregate Data Rate
AI	Artificial Intelligence	ANN	Artificial Neural Network
AR	Artificial Reality	B5G	Beyond 5G
BAIM	Big AI Models	CDR	Channel Data Rate
CR	Cognitive Radio	CSI	Channel State Information
CTDE	Centralized Training Decentralized Execution	D2D	Device-to-Device
DDPG	Deep Deterministic Policy Gradient	DNN	Deep Neural Networks
DQN	Deep Q Network	D-REX	Dialogue Relation Extraction with EXplanations
DRL	Deep Reinforcement Learning	DT	Delay Tolerant
eMBB	Enhance Mobile Broadband	ESS	Evolutionary Stable Strategies
FL	Federated Learning	GA	Genetic Algorithm
GenAI	Generative AI	GNNs	Graphic Neural Networks
GPTs	Generative Pretrained Transformers	HetNet	Heterogeneous Network
IIoT	Industrial Internet of Things	ILP	Integer Linear Programming
IoE	Internet of Everything	IoT	Internet of Things
IRS	Intelligent Reflecting Surfaces	LC	Latency Critical
LIME	Local Interpretable Model-Agnostic Explanations	LLMs	Large Language Models
LP	Linear Programming	LRU	Least Recently Used
LSTM	Long Short-Term Memory	LTE	Long Term Evolution
LWA	LTE-WLAN Aggregation	MAB	Multi-Armed Bandit
MARL	Multi-Agent Reinforcement Learning	MCS	Modulation and Coding Schemes
MEC	Multi-Access Edge Computing	MILP	Mixed-Integer Linear Programming
MIMO	Multiple Input Multiple Output	ML	Machine Learning
mMTC	Massive Machine Type Communication	MNO	Mobile Network Operator
M-QAM	M-ary Quadrature Amplitude Modulation	NDT	Network Digital Twin
NFV	Network Function Virtualization	NOMA	Non-orthogonal Multiple Access
NP-hard	Nondeterministic Polynomial-time hard	NR	New Radio
OFDM	Orthogonal Frequency Division Multiplexing	O-RAN	Open Radio Access Networks
PCA	Principal Component Analysis	PDPs	Partial Dependence Plots
PSO	Particle Swarm Optimization	PVDN	Prioritized Value-Decomposition Networks
QAGBRAS	QoE-Aware Bandwidth Resource Allocation Strategy	QoE	Quality of Experience
QoS	Quality of Service	RAN	Radio Access Network
RF	Radio Frequency	RL	Reinforcement Learning
SAGIN	Space–Air–Ground Integrated Network	SAGS	Space–Air–Ground–Sea
SC-FDMA	Single Carrier Frequency Division Multiple Access	SDN	Software Defined Networking
SEN	Super Edge Nodes	SHAP	SHAPley additive explanations
SL	Supervised Learning	SUs	Secondary Users
THz	Terahertz	TWDM-PON	Time and Wavelength Division Multiplexing Passive Optical Network
UAV	Unmanned Aerial Vehicle	UCB	Upper Confidence Bound
UHD	Ultra-High-Definition	UL	Unsupervised Learning
UMAP	UAV Movement and Power Allocation Policy	uRLLC	Ultra Reliable Low Latency Communication
UWGNN	Unrolled Weighted Graph Neural Network	V2N	Vehicle-to-Network
V2V	Vehicle to Vehicle	V2X	Vehicle-to-Everything
VLC	Visible Light Communication	VNF	Virtual Network Functions
VR	Virtual Reality	VU	Vehicular User
wBAIMs	wireless Big AI Models	WLAN	Wireless Local Area Networks
WMMSE	Weighted Minimum Mean Square Error	XAI	Explainable AI
XGBoost	eXtreme Gradient Boosting	XR	EXtended Reality

nature of SAGIN, caching is extended across satellites, UAVs, and terrestrial edge nodes. This geographic distribution reduces access delays and backhaul loads by storing popular content closer to end users in the integrated network environment. Additionally, semantic networking plays a vital role by optimizing caching and service delivery based on context extracted from service requests, allowing for more intelligent and efficient use of caching resources within the network [31].

Cross-domain collaboration of perception resources and computing resources in 6G involves integrating intelligent sensing and computation capabilities to enable more efficient and real-time data processing. This approach merges communication, sensing, and computation functions into unified platforms, where real-time sensory data collected from distributed sensors feed into AI-driven computation frameworks at the edge or cloud. Such collaboration enhances the accuracy, timeliness, and reliability of perceptual information while optimizing computational resource allocation. It aims to support complex applications like intelligent transportation and immersive environments by leveraging the synergy between perception and computing to improve decision-making, reduce latency, and ensure privacy and security [32]. SAGIN fosters multi-layer collaboration through AI-driven orchestration of heterogeneous resources. This collaboration enables seamless resource sharing and coordinated management across the space–air–ground continuum, optimizing network performance and resilience [6].

While sensing resources fall outside the 3C scope of this survey, the convergence of sensing and communication into a unified framework is rapidly gaining prominence under the paradigm of integrated sensing and communication (ISAC). ISAC enables the simultaneous use of shared spectrum and hardware for both radar sensing and wireless communication, improving spectral efficiency and supporting emerging applications such as autonomous driving, UAV navigation, and environmental monitoring [33]. Recent advances demonstrate that AI techniques, particularly DRL, are being applied to jointly allocate power and bandwidth between sensing and communication functions, balancing sensing accuracy against communication throughput [34]. As ISAC matures toward standardization, it was officially included in the scope of 3GPP 6G radio studies in 2025 [35]. Future extensions of the 3C framework may evolve toward a “3C+S” model that incorporates sensing as a fourth managed resource.

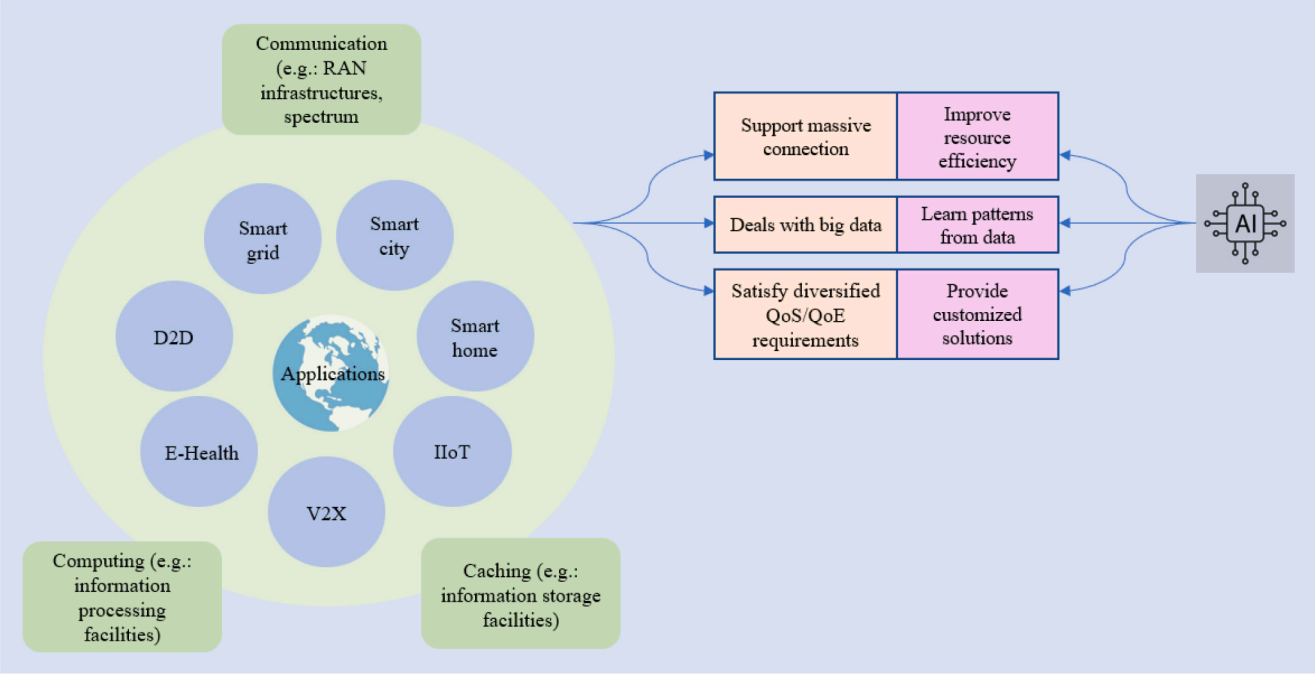


Fig. 1. Challenges in management of resources and possible AI solutions [16].

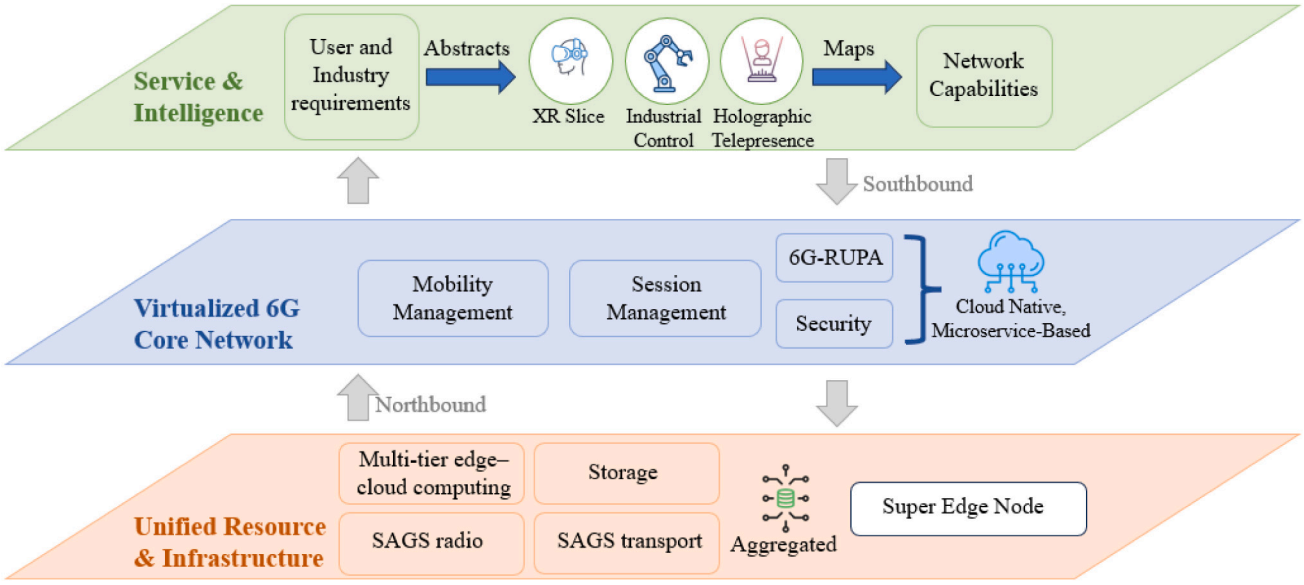


Fig. 2. Three-layer architecture for 6G networks.

1.3. Contribution and paper organization

This review examines peer-reviewed studies published between 2021 and 2026 that focus on AI-driven resource management in 5G and 6G wireless networks. To maintain both relevance and currency, particular emphasis is placed on works from 2023 to 2025 that explore the intersection of AI and the 3C resource framework. The surveyed literature includes journal articles, conference papers, and review studies published in English. Covered topics span AI-based resource allocation across the 3C framework, network slicing, CR, agentic AI models, XAI, and digital twins (DTs), reflecting the breadth of ongoing research in intelligent and trustworthy wireless systems.

We searched major databases, including IEEE Xplore, ACM Digital Library, ScienceDirect, SpringerLink, arXiv, and Google Scholar, using keywords like "artificial intelligence", "resource allocation", "6G", "wireless networks", and "machine learning". Boolean operators and relevant filters were applied. Initial screening of titles and abstracts was followed by full-text reviews of shortlisted papers for relevance. Multiple reviewers cross-checked selections, and key references were added through citation tracking. The review process followed a systematic evaluation approach similar to the framework outlined in [36], ensuring that the selected studies provided robust insights into AI-enabled resource allocation. This approach ensures a transparent, thorough, and replicable survey of recent AI applications in wireless resource management.

The main contributions of this work are:

- **AI-Empowered 3C Framework:** Presenting a 6G architecture for AI-driven resource allocation across the 3C continuum, integrating network slicing and CR.

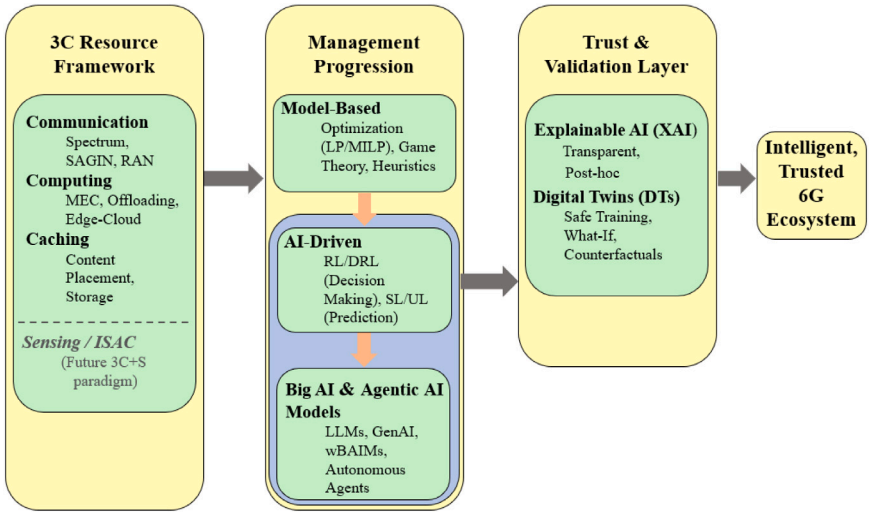


Fig. 3. A conceptual map of the surveyed field.

- **Comparative Methodologies:** Analyzing the shift from traditional model-based optimization to data-driven AI approaches, highlighting trade-offs in scalability and real-time execution.
- **Emerging agentic AI Paradigms:** Exploring the transition from task-specific agents to BAIM and agentic AI models for holistic, zero-touch network orchestration.
- **Trust and Validation Layer:** Integrating XAI with DTs to ensure transparent and safe network automation while minimizing explanation latency.
- **Future Outlook:** Identifying open challenges, such as cross-domain generalization and explainability-performance trade-offs, alongside actionable 6G research directions.

To clarify originality, this survey differs from historical generational reviews such as [37], which do not focus on AI-based resource allocation or systematic use of XAI. Earlier 5G-focused AI surveys mostly address radio resource management without explicitly covering 6G-specific resource types (3C, SAGIN/SAGS) or treating XAI as central. Our work advances by focusing on AI methods tailored to 6G use cases, jointly considering multiple resource types and introducing XAI as essential for transparent, trustworthy 6G resource management including monitoring and decision explanation.

The remainder of this paper is structured to reflect the methodological evolution required for intelligent 6G resource management, as visually summarized in the Conceptual Map (Fig. 3). Section 2 establishes the foundational 3C framework and the architectural context for resource allocation. Section 3 traces the progression of allocation strategies, contrasting traditional model-based approaches with adaptive AI-driven techniques. Building on this, Section 4 explores the shift toward Big AI and agentic AI models for autonomous network orchestration. To mitigate the opacity of these advanced models, Section 5 introduces XAI and DTs as a combined trust layer for verifying autonomous decisions. Finally, Section 6 concludes the paper and outlines future research directions.

2. Framework for resource management utilizing AI

The adoption of software-defined networking (SDN) and network function virtualization (NFV) has played a crucial role in making wireless communication management more adaptable, scalable, and resource-efficient [38]. The emergence of 6G is expected to significantly reshape how networks are designed and managed, transitioning from traditional entity-based architectures to function-oriented models customized for the needs of individual applications [39]. Mobile network operators (MNOs) can divide 3C resources into separate network slices. This capability allows them to deliver personalized slices tailored to users with varying QoS and QoE needs, or to offer different types of network slices to multiple tenants under diverse pricing models. Adapting to this new model calls for a restructured resource management system that operates across both operator and user domains. Engaging additional stakeholders and optimizing both resource allocation and network operations are critical priorities for operators. At the user level, optimizing the utilization of allocated resources is essential for effective resource management. This section provides an in-depth analysis of the resource management model illustrated in Fig. 4, highlighting the role of AI across multiple levels from network operators to end users. In this framework, "resource management" refers to the holistic orchestration of the entire network lifecycle, while "resource allocation" represents the dynamic mechanism responsible for assigning specific resources within that process.

2.1. Micro-operators and mobile network operators

MNOs and micro-operators are central to the emerging B5G and 6G ecosystem, where intelligent, adaptive network management becomes critical. While MNOs operate large-scale infrastructures spanning entire regions or countries, and micro-operators focus on localized networks tailored to specific venues or industrial applications, both face exponential growth in complexity stemming from ultra-dense connectivity, diverse service demands, and heterogeneous spectrum usage. In this environment, AI is no longer optional but fundamental to sustaining efficiency, resilience, and service quality [40,41].

AI-driven automation enables MNOs to handle massive volumes of network data generated across layers of radio, transport, and core systems in real time. Traditional rule-based management cannot scale to the diversity and dynamism of B5G/6G scenarios, where resource demands fluctuate instantaneously with user mobility, traffic patterns, and environmental conditions. ML algorithms can predict network loads, optimize spectrum allocation, and coordinate distributed edge resources allowing MNOs to maintain high-quality, low-latency connectivity while reducing operational costs and energy consumption [40,41]. In the context of the Open Radio Access Network (O-RAN), this automation is increasingly powered by cloud-native multivariate probabilistic forecasting [42]. By leveraging probabilistic models to predict resource demands, operators can achieve more resilient and next-generation resource allocation that accounts for the inherent uncertainty in 6G traffic patterns and user mobility.

For micro-operators, AI supports localized intelligence by autonomously tuning network parameters to meet specialized performance goals, such as ultra-reliable low-latency communications

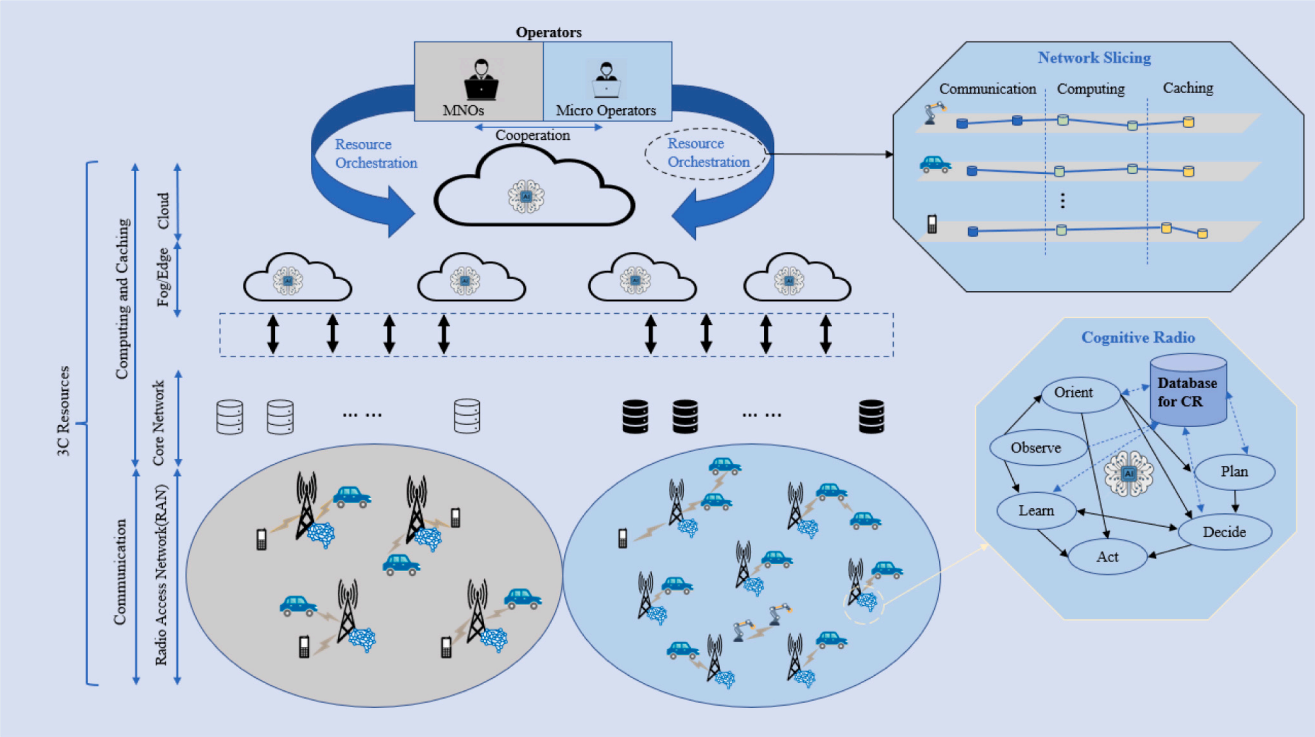


Fig. 4. 6G architectural framework [16].

(URLLC) in factories or high-throughput coverage in dense venues. These operators often lack the human and computational resources of MNOs, making AI essential for adaptive configuration, fault prediction, and real-time interference mitigation. Through reinforcement learning (RL) and digital twin technologies, micro-operators can continuously optimize deployment strategies and network slicing arrangements to maximize performance within limited footprints [40].

The partnership between MNOs and micro-operators further amplifies the need for AI-based coordination. The integration of heterogeneous networks, the dynamic sharing of spectrum, and the execution of seamless handovers between public and private domains necessitate intelligent, cross-entity orchestration. AI-powered control systems can automate cross-operator negotiations, balance loads across shared resources, and predict congestion before it occurs, ensuring uninterrupted service continuity. This level of collaboration cannot be achieved through static configurations or manual oversight [40,41].

In summary, the coexistence and interdependence of MNOs and micro-operators in the B5G/6G era demand AI-driven intelligence across every layer of network operation from spectrum planning to service delivery. As networks evolve toward greater autonomy and contextual awareness, AI becomes the core enabler of efficiency, adaptability, and scalability in an increasingly complex and interlinked mobile ecosystem [40,41].

2.2. Network slicing technologies within 6G systems

Network slicing is a networking concept in which a single physical infrastructure is partitioned into multiple logical networks, called slices, each tailored to a specific service or tenant with its own performance and security requirements. By leveraging softwarization and virtualization, operators can dynamically create, modify, and remove these slices. This flexibility allows diverse services, such as massive IoT and enhanced mobile broadband, to run simultaneously on the same underlying hardware. Crucially, each slice maintains strict logical isolation to enforce its specific resource allocation and QoS guarantees [43].

Network slicing in 6G relies on AI because human designed control cannot handle the scale and variability of future slices, users, and services. AI enables autonomous, predictive slicing that jointly optimizes QoS/QoE, cost, and energy across shared RAN and core resources. With rapidly changing traffic, mobility, and service mixes, rule-based methods are intractable and heuristics insufficient for exploiting 3C resources under strict QoS/SLA constraints [44]. To address this, recent advancements utilize ML for dynamic slicing reconfiguration by specifically forecasting computing capacity within virtualized networks [45]. These models allow for proactive resource adjustment before violations occur, ensuring that the computing pillar of the 3C framework remains resilient under fluctuating workloads.

AI-native slicing integrates learning agents into the SDN/NFV control stack so slice admission, scaling, routing, and RAN configuration rely on online predictions rather than static templates. In the RAN, AI tunes the isolation–efficiency trade-off as slices share physical and MAC resources (see Fig. 5). At the service layer, DRL and evolutionary methods learn SLA-aware admission and pricing policies that balance revenue, cost, and QoE across dynamic tenant and traffic conditions [46].

Lifecycle management for thousands of slices requires zero-touch operation. AI/ML agents enable proactive scaling, cross-domain coordination, and self-healing through multi-time-scale prediction and control. At the infrastructure level, AI-based schedulers allocate spectrum, power, antennas, computing, and cache for diverse eMBB, URLLC, mMTC, and XR/AI services using contextual data such as channel state and mobility [47].

As 6G must meet stringent latency, reliability, connectivity, and energy targets in dynamic environments, AI learns trade-offs between throughput, delay, energy, and isolation that static algorithms cannot. XAI ensures interpretability, compliance, and trust in safety-critical domains like vehicular and industrial networks [48]. Since many 6G services are AI-centric such as collaborative robotics and edge intelligence AI-driven slicing jointly allocates communication and computation to meet end-to-end QoS at low cost and energy. This makes 6G slicing an AI-native, closed-loop system delivering scalable, autonomous, and trustworthy operation [49].

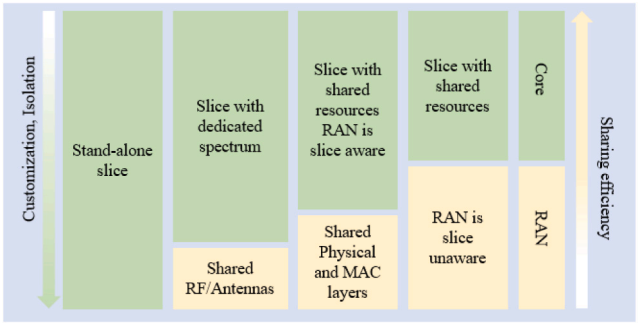


Fig. 5. RAN slice multiplexing methods [16].

2.3. Cognitive radio in 6G resource allocation

CR acts as a critical enabler within the communication pillar of the 3C framework. Unlike independent resource types, CR is specifically a spectrum management technique designed to dynamically utilize available radio frequency bands. Hence, resource allocation in CR belongs to a subcategory of communication resources, particularly tailored to the unique requirements of spectrum sharing and interference management. The goal of CR in 6G networks is to maximize spectrum efficiency by allowing secondary users (SUs) to opportunistically access frequency bands without interfering with primary users [50]. As 6G systems evolve toward hyper-dense, heterogeneous, and highly dynamic environments, traditional rule-based or static resource allocation approaches become insufficient. The complexity of real-time spectrum sensing, interference avoidance, and adaptive reconfiguration calls for intelligent, data-driven methods making AI indispensable in CR resource management.

AI plays a pivotal role in every stage of the cognitive cycle: spectrum sensing, decision, sharing, and mobility [51]. Instead of relying on fixed thresholds or handcrafted algorithms, AI enables CR to learn from ambient radio conditions and user behavior, predict spectrum availability, and proactively allocate resources. This predictive and adaptive intelligence is especially critical in 6G, where latency, reliability, and energy efficiency demands require CR systems to make near-instantaneous, context-aware decisions.

The sheer number of devices competing for spectrum in 6G leads to unprecedented interference and coordination challenges [52]. AI-based models, such as deep learning and RL, allow CR to autonomously detect patterns in spectrum usage, optimize transmission parameters, and dynamically balance load across multiple frequency bands. Unlike conventional approaches, AI-driven CR systems can generalize from experience, handle incomplete information, and continuously refine their strategies over time.

Furthermore, the decision-making landscape in CR is highly complex, encompassing multi-dimensional variables such as RF bands, antenna configurations, user mobility, and environmental factors. This complexity aligns naturally with AI's strength in managing high-dimensional data and uncertain environments. Techniques inspired by the Multi-Armed Bandit (MAB) framework, for instance, enable SUs to explore and exploit spectrum opportunities without explicit coordination, improving long-term spectrum efficiency [53,54]. AI-based fusion centers can also dynamically determine detection thresholds, adapt to non-stationary noise conditions, and enhance spectrum sensing accuracy.

Therefore, AI is not merely an enhancement to CR; it is a foundational enabler that allows CR to achieve self-learning, self-optimizing, and self-healing capabilities required for intelligent, adaptive resource allocation in 6G networks. Through AI, CR evolves from a reactive system to a proactive, context-aware agent capable of managing spectrum scarcity with unprecedented precision and efficiency.

3. Resource allocation methodologies

This section examines the methodological evolution essential for effective 6G resource management. Section 3.1 establishes the theoretical foundations grounded in traditional optimization techniques and heuristic methods. These approaches are revisited primarily to expose critical limitations particularly in adaptability and scalability that motivate the AI-driven strategies explored in Section 3.2. Through this comparison, we underscore the necessity of a paradigm shift toward data-driven AI methodologies.

3.1. Limitations of model-based approaches for resource allocation

Model-based resource allocation relies on mathematical optimization, game theory, and heuristics, but these frameworks struggle with the dynamic, nonlinear complexities of modern wireless networks. Mathematical optimization methods like linear programming (LP), integer linear programming (ILP), and mixed-integer linear programming (MILP) often face NP-hard computational challenges and scalability issues in large-scale 5G and beyond deployments. Heuristics provide practical approximations but sacrifice optimality, while game theory assumes rational agents in unpredictable environments [55,56].

3.1.1. Mathematical optimization models

Mathematical optimization models, including LP, ILP, and MILP, suffer from high computational complexity due to their NP-hard nature, making them intractable for real-time resource allocation in dynamic 5G networks [55]. LP formulations in [57] assume static parameters and linear constraints, failing to capture nonlinear channel variations, interference, and mobility, which limits applicability to real-world HetNets with time-varying traffic. Scalability degrades in large networks, necessitating heuristics that yield only near-optimal solutions despite tractable small-scale performance [55].

ILP models in TWDM-PON cell-free massive MIMO systems [58] grapple with exponential complexity from discrete variables like wavelengths and resource blocks, restricting scalability to small networks under static traffic assumptions. Fronthaul capacity constraints exacerbate issues, as dynamic conditions demand real-time adaptations beyond solver capabilities like Gurobi. Heuristic approximations improve feasibility but introduce sub-optimality gaps in resource efficiency for 6G-scale deployments.

MILP approaches for LoRa networks [55] handle mixed variables but impose static constraints ill-suited for fluctuating IoT environments, leading to poor adaptability and packet collision risks. Solver-genetic algorithm hybrids reduce energy use yet falter in large-scale, real-time scenarios due to prohibitive computation times and sensitivity to hyperparameters. Overall, these models highlight the need for AI-driven alternatives to overcome tractability limits in beyond-5G networks.

3.1.2. Game theory-based models

Game-theoretic models for resource allocation in wireless networks assume rational agents reaching Nash equilibria, but struggle with convergence delays, sensitivity to imperfect information, and scalability in dynamic 5G and beyond environments [59]. Non-cooperative frameworks like those in [60] for SC-FDMA heterogeneous networks rely on distributed learning algorithms such as self-adaptive logit, which incur high signaling overhead and fail to guarantee global optimality amid co-tier and cross-tier interference. These approaches overlook latency, fairness, and real-time validation, limiting deployment in large-scale, volatile settings [56].

Cooperative game models for fog-assisted 5G demand extensive information exchange and accurate parameter estimation for incentive bargaining. However, this introduces overhead that degrades performance under heterogeneous node capabilities and dynamic loads [61].

Dependency on trust mechanisms increases vulnerability to misbehavior, while lack of real-world testing exposes gaps in load balancing sustainability. Such centralized coordination contradicts the decentralized ethos needed for scalable networks.

Evolutionary games in [62] for 6G use replicator dynamics and ESS, yet suffer computational complexity from payoff matrix calculations and sensitivity to initial conditions, hindering fast convergence in rapidly changing conditions. Stability assumes perfect strategy evolution, unrealistic amid node mobility and imperfect observations, necessitating broader validation beyond simulations.

Stackelberg models like QAGBRAS in [63] for B5G UHD video impose hierarchical leader-follower structures with high real-time complexity, struggling with multi-objective optimization under congestion. Assumptions of perfect information and static hierarchies falter in fluid environments, yielding suboptimal QoE and revenue balances without extensive real-world trials. Overall, game theory requires extensive information and complex equilibrium computations, which limits its practicality for beyond-5G resource challenges [64].

3.1.3. Heuristic models

Heuristic models, including classical greedy and local search methods like tabu search, sacrifice global optimality for computational speed but often trap in local optima and fail to scale in dynamic 6G networks [65]. These problem-specific approximations overlook nonlinear interdependencies and require extensive tuning, limiting robustness amid VLC clipping distortion or V2X mobility.

Greedy bipartite graph algorithms in multi-tier VLC networks [65] prioritize local QoE gains via iterative power and M-QAM allocation, yet miss global optima due to myopic decisions and inaccurate interference modeling in novel architectures. Scalability falters under dynamic user mobility and multi-tier heterogeneity, demanding real-world validation beyond controlled simulations.

Tabu search in multi-access edge computing (MEC)-assisted V2X networks [66] decomposes partial offloading into subproblems for delay-energy minimization, but simulation reliance ignores real-world dynamics like dense traffic fluctuations. Careful hyper-parameter tuning proves infeasible in real-time, high-mobility settings, while hybrid complexity hinders deployment scalability. Overall, heuristics' sub-optimality and brittleness underscore the need for adaptive AI alternatives in beyond-5G resource challenges.

The model-based resource allocation techniques summarized in Table 2 loose momentum in future wireless networks due to their rigid reliance on predefined mathematical models and static parameters. These approaches struggle with nonlinear dynamics, time-varying interference, and user mobility inherent in 6G environments, yielding suboptimal performance under uncertain conditions. Computational intensity escalates with network scale, rendering real-time decision-making infeasible and exposing scalability bottlenecks in large deployments. These limitations, including NP-hard complexity, convergence delays, local optima trapping, and imperfect information handling, underscore the inadequacy of traditional methods. Consequently, adaptive AI-driven alternatives are necessary for robust, data-informed resource management.

3.2. AI-based resource allocation

This section analyzes how data-driven paradigms optimize the 3C resources. Unlike model-based methods, AI can learn complex correlations among these resources, such as the trade-off between communication bandwidth and computing latency, without relying on fixed mathematical assumptions.

In wireless networks, AI-driven methods enhance coverage and performance while balancing accuracy and tractability. By inferring resource allocation directly from data, AI reduces reliance on traditional statistical models and offers efficient solutions to NP-hard problems. It also accommodates more parameters in learning and decision-making, enabling holistic optimization of multi-objective problems without decomposition [67].

AI-based models for network optimization are generally categorized by learning paradigm and objective: supervised learning (SL), unsupervised learning (UL), RL, and bio-inspired learning.

3.2.1. Supervised learning models

SL is primarily used to predict resource demands and channel states, enabling proactive resource allocation across the communication and computing domains. By mapping historical data to optimal outputs, SL models approximate complex optimization functions for spectrum management (communication) and task latency estimation (computing). The goal is to improve key performance indicators such as network throughput, energy efficiency, and latency. Common SL techniques in this field include artificial neural networks (ANNs), graph neural networks (GNNs), and deep neural networks (DNNs) [68].

Research indicates that ANNs are increasingly employed in resource allocation for telecommunication networks, especially within the context of 5G and beyond. Recent studies demonstrate this versatility across diverse architectures; for instance, ANNs have been applied to heterogeneous networks for joint mobile relay selection [69], as well as to optical wireless networks for laser-based resource allocation [70] and cooperative rate maximization [71]. Narrowing focus to the computing domain, research [72] uses ANNs to predict edge task latency, enabling joint optimization of computing and communication resources for delay-sensitive applications. The ANN models estimate three key metrics: the blocking probabilities for latency-critical tasks, the blocking probabilities for delay-tolerant tasks, and the preemption probability for delay-tolerant tasks. This final metric indicates the risk of a task being interrupted by higher-priority traffic. The ANN-assisted allocation scheme jointly optimizes these metrics to ensure ultra-reliable edge computing. It minimizes total blocking probability while maintaining QoS by keeping preemption and blocking rates below defined thresholds. This balance effectively manages spectrum and storage resources to meet growing data exchange demands among edge nodes. By adapting resource use via ANN predictions, the method enhances network responsiveness, achieving low latency and high throughput. However, the effectiveness of such models depends on accurate training data and robust design, while challenges remain in reducing computation overhead and integrating with existing systems. Addressing these computational constraints, specialized lightweight architectures like the AERO framework offer a viable alternative [73]. By utilizing a specialized ANN with fewer than 1000 parameters, AERO achieves adaptive edge–cloud orchestration while minimizing the computational footprint. This shift toward ultra-compact ANNs ensures that high-accuracy forecasting remains feasible even on resource-constrained edge devices, avoiding the high latency associated with traditional ANN models.

Complementing this focus on efficiency, recent research also targets the challenge of model adaptability across the heterogeneous compute continuum. For instance, [74] introduce OmniFORE, an attention-driven AI framework for workload forecasting. Functioning as a hybrid ANN, this model integrates two distinct architectures: convolutional neural networks for temporal clustering and Attention-Based Transformers (Informers) for forecasting. By leveraging these combined mechanisms, the approach enhances model generalization, enabling accurate predictions across diverse edge and cloud environments without the need for extensive retraining specific to each node.

GNNs have emerged as a powerful tool for modeling the irregular and dynamic topologies characteristic of 6G networks. Recent literature demonstrates their versatility across diverse resource management scenarios. These applications include SL frameworks for IoT networks [75] and interference-limited multi-channel systems [76], as well as advanced DRL models for SAGIN [77] and THz-band space communications [78]. Highlighting this capability in terrestrial interference management, research [79] employs an unrolled weighted

Table 2
Model-based resource allocation methods.

Model category	Model	Application method	Limitations/Challenges
Mathematical	LP	Joint allocation of radio and transport resources in future HetNets [57]	Assumes static parameters and struggles with nonlinear, dynamic wireless conditions
	ILP	Joint resource allocation in TWDM-PON-enabled cell-free massive MIMO to minimize fronthaul usage [58]	Model assumes static traffic and conditions
	MILP	Minimize energy consumption while maximizing user fairness in LoRa networks [55]	Static constraints reduce adaptability to real-time network changes
Game theory	Non-cooperative games	Manage co-tier, cross-tier, and inter-cell interference in heterogeneous SC-FDMA uplink networks using a non-cooperative game approach [60]	High learning overhead, limited large-scale validation, and focus on throughput over latency or fairness
	Cooperative games	Enable resource sharing among fog nodes [61]	High information exchange overhead, reliance on accurate parameters, and limited validation in dynamic large-scale environments
	Evolutionary games	Adaptive and efficient resource distribution under varying network conditions [62]	High computational complexity, sensitivity to payoff accuracy, and limited real-world testing
	Stackelberg games	Bandwidth allocation for UHD video in B5G networks [63]	Real-time complexity, traffic variability, and limited real-world testing
Classic heuristic	Greedy algorithms	Resource allocation in multi-tier VLC networks [65]	May miss global optima, relies on accurate interference modeling, and faces scalability and adaptability issues in multi-tier VLC architectures
	Local search algorithms	Overcome full offloading limitations in MEC for V2X networks [66]	Simulation-based validation may miss real-world dynamics; scalability, real-time performance, and parameter tuning need further testing

graph neural network (UWGNN) to optimize wireless interference management through the weighted minimum mean square error (WMMSE) framework, reducing transmission latency for remote computing tasks. UWGNN is a deep learning model for efficient resource allocation in wireless networks, maximizing the weighted sum rate in device-to-device (D2D) communications by optimizing transmission power. It achieves millisecond-level responsiveness three to four orders faster than conventional WMMSE while maintaining scalability, generalization across topologies, and low model complexity. The network is modeled as a directed graph, where each node represents a D2D pair characterized by features such as channel gain, priority weight, and transmission power. The edges between these nodes capture the interference, which is structured via an adjacency matrix. Using Rayleigh or Rician channel data, UWGNN outputs optimized transmission power through its final submodule, applying a sigmoid function to keep power within limits. The model generalizes well without retraining, enabling real-time, low-latency resource management. Although robust and scalable, deployment requires attention to dynamic conditions and system integration.

By leveraging their ability to approximate complex non-linear functions, DNNs can learn the intricate relationships between network states and optimal resource allocation decisions. For example, the authors in [80] introduce ALCOR (a deep learning–based resource allocator), which manages resource allocation in communication systems with time-varying user demands, such as data rate and latency requirements. Embedding DNNs in an iterative optimization process, ALCOR adapts to changing constraints without retraining, improving efficiency. It aims to maximize the instantaneous sum utility across active users while satisfying each user's expected demand. The policy network processes time-varying inputs such as channel coefficients, interference, queue lengths, transmit power history, and data rates. It then outputs optimal resource allocations, which may include transmit power, subchannels, time slots, or beamforming directions. Coupling a pre-trained unconstrained resource allocation policy with a time-sharing algorithm, ALCOR selects active users to meet long-term demands while maintaining overall efficiency. Modeling user demands as stochastic variables, it ensures convergence and performs better than traditional methods, even under infeasible conditions. While adaptive, scalable, and theoretically sound, its assumptions on demand and channel dynamics and the latency of its iterative process may limit practical use.

Within the caching domain, SL plays a foundational role through content popularity prediction, which enables proactive cache placement at the network edge. Rather than reactively storing content upon first request, SL models learn temporal and contextual patterns from historical request data to forecast which content will be demanded in the near future. Long short-term memory (LSTM) networks have emerged as a particularly effective architecture for this task due to their ability to capture long-range temporal dependencies in content request sequences. For example, [81] propose a proactive edge caching framework that uses an LSTM network to predict time-varying content popularity at SBSs. The model processes historical request logs, temporal features, and base station load indicators to predict a popularity ranking across the content catalogue. This ranking directly drives cache placement decisions, enabling the system to pre-fetch top-ranked items to the edge cache before users request them. The objective is to maximize the cache hit ratio while operating within the finite storage capacity of edge nodes. Simulation results demonstrate that LSTM-based proactive caching significantly outperforms reactive baselines such as least recently used (LRU) in terms of both cache hit rate and content retrieval latency.

Extending this predict-then-cache paradigm to privacy-sensitive environments, [82] introduce an FL approach for popularity prediction, where each user trains a local neural network on its own request history without uploading raw data to the base station. The aggregated model provides base-station-level popularity estimates that drive caching decisions while preserving user privacy. This is particularly relevant for 6G networks where edge caching must comply with data protection requirements. However, SL-based caching approaches share several common limitations: they require sufficient historical data to train accurate models, face a cold-start problem for newly published content with no request history, and cannot easily adapt to abrupt shifts in content popularity caused by viral events. These limitations motivate the use of RL for caching (Section 3.2.3), where the agent can learn adaptive placement policies through online interaction rather than relying on static prediction models.

3.2.2. Unsupervised learning models

UL is critical for structuring unstructured network data, particularly for organizing users and content in the communication and caching domains. Clustering algorithms like K-means are widely used to group

users spatially to mitigate interference (communication) or to identify popular content patterns for proactive storage (caching). These approaches enable adaptive, low-complexity solutions for dynamic network environments such as CR, NOMA IoT, and D2D networks [83].

Resource allocation approaches using K-means clustering in telecommunication networks focus on grouping users or network nodes into clusters to optimize the allocation of limited resources such as bandwidth, spectrum, power, or time slots. K-means helps reduce complexity by effectively managing resource distribution based on spatial or usage patterns. To optimize communication resources in UAV-assisted networks, [84] uses K-means clustering to spatially group users, reducing UAV–user distances and improving channel quality. By integrating UAV, millimeter wave (mmWave) MIMO, and NOMA, the framework enhances data rates and lowers transmission delays. K-means addresses the UAV deployment sub-problem by clustering users to identify optimal UAV positions, minimizing average user distance and thus boosting channel gains and throughput. The algorithm accepts user coordinates, the number of UAVs, and initial center points as inputs. It then outputs optimized UAV coordinates, user associations, cluster sizes, and pairings that support effective successive interference cancellation. With computational complexity $O(UKt_n)$, where U, K, t, and n denote UAVs, users, iterations, and data dimensions, K-means-based deployment still achieves lower latency than random placement. The full model significantly reduces delay and improves throughput by jointly optimizing three key areas. Specifically, it handles UAV deployment via K-means, hybrid beamforming using particle swarm and zero-forcing, and power allocation via a genetic algorithm (GA). However, K-means' sensitivity to initial centers, its spherical cluster assumption, and limited adaptability to user mobility and channel dynamics pose challenges for real-time application. Future work should enhance scalability, mobility management, and caching efficiency to improve practicality in dynamic networks.

The paper [85] presents a clustering-based D2D resource allocation scheme using the K-means algorithm to reduce interference and computational complexity. By defining restricted D2D communication and reuse zones, the method minimizes overlapping transmissions and manages interference effectively. Spatial user clustering simplifies resource allocation, enhances spectral efficiency, and ensures QoS compliance. The model aims to maximize total system throughput in a D2D-enabled cellular network while maintaining fairness and spectrum efficiency. It minimizes interference among D2D pairs, between D2D transmitters and the base station, and between cellular users and D2D receivers, ensuring all users meet minimum SINR thresholds. Using inputs such as user distribution, channel data, and system constraints on power, resources, and interference, the algorithm outputs optimized D2D clusters, an allocation matrix for resource blocks, and transmission power levels. Performance metrics, including total capacity and average SINR, confirm improved efficiency and fairness. However, K-means' sensitivity to initial clusters, its assumption of spherical distributions, and limited adaptability to user mobility restrict scalability in dynamic environments. Overall, the scheme offers an effective and practical approach for D2D resource management but would benefit from enhanced mobility handling and real-time adaptability.

In the caching domain, K-means clustering identifies localized content demand, recognizing that popularity varies by geography and demographics. Clustering base stations by historical requests yields local popularity distributions that significantly improve cache hit rates over global models. For example, the proactive caching scheme in [86] clusters small-cell base stations (SBSs) based on the cosine similarity of their historical request vectors. Within each cluster, local content popularity is estimated via linear regression. Cache placement is then optimized using stochastic geometry and convex optimization (Lagrangian relaxation and KKT conditions) to output a customized caching policy. Real-world evaluations confirm this localized approach outperforms global schemes. However, static K-means clusters fail to adapt to temporal shifts in user behavior, highlighting the need for dynamic re-clustering or RL integrations in future work.

3.2.3. Reinforcement learning models

RL acts as the core decision-making engine for dynamic resource orchestration, especially when communication and computing resources must be allocated sequentially under uncertainty. In telecommunication networks, RL-based approaches dynamically optimize resources such as spectrum, power, computational capacity, and bandwidth to adapt to fluctuating demands and complex network environments. RL methods learn effective allocation policies over time by interacting with the network environment, adapting to real-time conditions without requiring explicit, fixed models of network dynamics. At its core, an RL framework consists of an agent (the decision-maker, e.g., a base station), an environment (the network), a set of states (network conditions), a set of actions (resource allocation decisions), and a reward signal (a measure of performance). The agent learns a policy, a mapping from states to actions, that maximizes its cumulative reward over time. Several RL algorithms have been prominently featured in the literature for network resource allocation, such as Q-learning, DRL, policy gradient methods and multi-agent reinforcement learning (MARL) models [87].

DRL models are widely used to capture spatial and temporal variations in user demand and network conditions. By adapting to these dynamic patterns, DRL can efficiently allocate computational and radio resources across diverse high-mobility environments. Recent studies illustrate this capability in terrestrial systems, such as optimizing age of information (AoI) in RIS-aided internet of vehicles (IoV) [88], managing joint computation-communication resources in railway IoT [89], and enabling decentralized allocation in vehicular networks via federated multi-agent learning [90]. Extending this adaptability to the aerial domain, research [91] introduces the UAV Movement and Power Allocation Policy (UMAP), a DRL-based framework that jointly optimizes communication (coverage and power) and computing (MEC user association) resources. UMAP uses DRL to iteratively optimize UAV trajectories and user associations, while transmit power is obtained through a closed-form solution. The model minimizes the weighted sum of system latency and energy consumption while ensuring QoS under constraints such as task latency, transmit power limits, UAV capacity, flight speed, battery life, and collision avoidance. Two DRL agents coordinate the process. First, a UAV Movement Agent determines flight distance and direction based on UAV positions, user locations, and power allocation data. Second, a Mobile User Association Agent selects offloading targets using channel gains between users, UAVs, and the base station. This joint optimization reduces latency and energy consumption, outperforming traditional methods in efficiency and coverage. While UMAP effectively balances performance and complexity, it depends on accurate network data and incurs high training costs, limiting real-time applicability. Nonetheless, it represents a promising and scalable solution for UAV-assisted MEC systems, with room for improvement in adaptability and deployment efficiency.

Q-learning and its deep or distributed variants remain foundational for adaptive resource allocation, with recent innovations addressing diverse 6G constraints. Implementations range from Deep Q-learning for managing interference under strict outage constraints [92] and lightweight distributed frameworks for energy-efficient LoRa networks [93], to advanced scalable solutions utilizing "digital cousins" for ensemble learning [94] and multi-agent mixed environments for decentralized optimization [95]. In the specific context of multi-numerology network slicing, classic Q-learning is used in beyond 5G networks to exploit long term evolution-wireless local area network (LTE-WLAN) aggregation for balancing resource block allocation [96]. This study proposes a joint Q-learning framework to optimize resource allocation in beyond 5G networks with multi-numerology network slicing, where slices use different subcarrier spacings to support varied applications. By incorporating LTE-WLAN aggregation (LWA), the model combines heterogeneous resources to enhance capacity and efficiency. Its goal is to maximize user satisfaction by jointly optimizing

resource allocation and numerology configuration across slices, ensuring SLA compliance high throughput for eMBB and low latency for URLLC. The environment state includes the resource block ratio and numerology value (0–4), while the action space allows the agent to adjust these parameters to learn an adaptive allocation strategy. The Q-learning agent dynamically balances resources among slices, improving QoS adaptability and system efficiency. Although effective, the approach faces scalability and training challenges due to large state–action spaces and LTE-WLAN integration limits. Overall, it provides a flexible, data-driven solution for dynamic resource management in beyond 5G networks.

Policy gradient methods, especially deep deterministic policy gradient (DDPG), are increasingly used in resource allocation for telecommunication networks to handle dynamic, complex, and high-dimensional optimization problems. For instance, [87] introduces a DDPG-based frequency subband allocation method for 6G aerial vehicular networks using user-centric, cell-free massive MIMO UAVs. The RL agent optimizes resource allocation by balancing interference, spectral efficiency, and fairness under bandwidth constraints. The state includes environmental factors such as UAV positions, interference levels, QoS requirements, and channel characteristics influenced by UAV mobility and altitude, along with available CSI. The action, produced by the actor network, generates subband assignment policies by constructing an allocation matrix that determines user–subband mappings. The reward function jointly maximizes spectral efficiency, ensures fairness through Jain's Index, and minimizes interference while satisfying power and QoS constraints. This method outperforms traditional bio-inspired heuristics in dynamic networks but requires extensive training data, high computational resources, and may face slower convergence in fast-changing conditions. While effective, improving scalability and real-world adaptability remains a key challenge.

MARL has been widely applied for resource allocation in telecommunication networks to cope with their dynamic, complex, and distributed nature. In such applications, network nodes are modeled as independent agents that learn to cooperatively allocate resources. MARL enables decentralized decision-making, adapting to changing traffic demands and channel conditions while optimizing key metrics such as network throughput, energy efficiency, latency, and QoS. For instance, [97] tackles energy-efficient resource allocation in heterogeneous MEC networks by optimizing edge server computing under nonlinear energy constraints. It introduces a fully decentralized multi-agent deep reinforcement learning (MADRL) framework with a multi-actor shared-critic design, regional training, and distributed execution. Rather than relying on global information, each agent in the Het-MEC network depends solely on local observations. By utilizing data from previous steps, such as CSI, past resource allocations, and transmission rates, the agents are able to make delay-tolerant, predictive decisions. Agents jointly decide on subcarrier assignment and power control within predefined limits. The model's goal is to maximize long-term energy efficiency, measured as the ratio of total transmission rate to total power consumption, ensuring stable and sustainable service. A scenario-oriented reward function allows decentralized learning, improving training stability and reducing inter-agent communication. Simulation results show superior convergence speed, scalability, and energy efficiency over baseline methods. Despite advantages in flexibility and practicality, challenges remain in training complexity, agent coordination, and real-world deployment.

While the preceding DRL examples focus on communication and computing resources, RL is equally critical for the caching pillar, where the agent must decide what content to cache, when to evict, and how to coordinate caching with offloading decisions. Authors of [98] propose an LSTM-DDPG framework that jointly optimizes computation offloading and edge service caching. The LSTM component captures long-term environmental state characteristics and memory information to model temporal dependencies in task arrivals and content requests. The DDPG component then uses this enriched state representation to make continuous offloading and caching decisions. The state space consists of the total cost of all users in the previous time slot, the remaining computational resources of the edge server, the remaining storage resources of the edge server, and the remaining channel bandwidth. The action space encompasses the offloading decision vector (local, edge, or cloud), the computational resource allocation vector, and the channel bandwidth allocation vector. The reward function is designed to minimize the long-term average system overhead, which is defined as a weighted combination of service latency and energy consumption. To maximize collaborative advantages across distributed edge regions, the framework incorporates a distributed edge caching algorithm (DECA) that uses a distributed hash table network and a cache weight factor-based eviction strategy to improve the overall cache hit rate. Simulation results demonstrate that the combined LSTM-DDPG with DECA outperforms standalone DRL offloading algorithms and conventional caching strategies in terms of system overhead, cache hit rate, and backhaul load reduction.

In multi-server edge environments, caching decisions at one edge node affect neighboring nodes through content redundancy and user mobility patterns, making cooperative caching inherently a multi-agent problem. [99] propose a content caching framework for vehicular edge computing using multi-agent graph attention RL. Each vehicular user (VU) acts as an independent agent observing its local cache state, content request history, and traffic flow. A graph attention mechanism captures relationships among VUs to learn spatial–temporal patterns. When completing a task, agents decide whether to cache the result and select a target service node (road side unit (RSU) or VU). The system minimizes total content access latency across all RSUs using centralized-training-decentralized-execution (CTDE). This approach outperforms independent DQN-based and heuristic baselines by exploiting spatial–temporal correlations in content demand. However, the graph attention mechanism's computational complexity limits scalability to very large networks with thousands of edge nodes.

3.2.4. Bio-inspired algorithms

Resource allocation in telecommunication networks using bio-inspired algorithms is an active research domain that leverages natural processes such as genetics, swarm intelligence, and biological evolution to optimize the distribution of critical resources like bandwidth, power, and computational capacity. Algorithms such as GAs, particle swarm optimization (PSO), ant colony optimization (ACO) are widely employed due to their ability to efficiently navigate complex, and dynamic optimization landscapes. These methods are particularly effective in avoiding local optima and adapting to fluctuating network conditions, making them ideal for modern telecommunication systems characterized by high mobility and variable demand.

By simulating evolutionary processes, GA can improve system throughput, fairness, and resource usage while reducing interference and energy consumption. These methods often outperform conventional approaches, especially in environments with heterogeneous resources and dynamic user demands. For example, The article [100] introduces MOMA, a GA-based framework for optimizing computing resources in containerized, heterogeneous clouds. The model requires several data types to initialize and evaluate the environment, including system entities (clusters, hosts, and microservices), relational matrices (distance and interaction matrices capturing node separation and microservice communication), and performance metrics such as CPU and memory usage, request rate, and execution time. It also uses file metadata (type and size) for data placement and GA parameters population size of 200, mutation probability of 0.3, and 25,000 iterations to guide its operation. MOMA outputs an optimized cloud configuration, providing a Pareto front of optimal trade-offs, microservice placements, balanced workload distribution, and server categorization based on metadata. Framing resource management as a multi-objective optimization problem, it seeks to maximize utilization, minimize communication overhead, improve reliability, balance

load, and reduce task execution and retrieval times. Overall, MOMA enhances load distribution, energy efficiency, and system stability, outperforming traditional approaches. Its strength lies in optimizing multiple objectives simultaneously, making it well-suited for dynamic cloud environments. However, its effectiveness depends on accurate workload modeling and GA parameter tuning, while scalability and computation overhead remain challenges. Despite these, MOMA is a promising step toward intelligent, real-world resource optimization.

The study in [101] proposes a PSO based algorithm to optimize communication resource allocation in mobile WiMAX networks, focusing on efficient distribution of subcarriers and Orthogonal OFDM symbols for multicast traffic. The WiMAX cell is modeled as seven concentric layers with distinct modulation and coding schemes (MCS) based on user distance from the base station, user distribution, and initial time–frequency resource assignments to multicast subgroups. In the PSO framework, each particle represents a candidate mapping of OFDMA resources to multicast groups. Particle positions and velocities are iteratively updated under WiMAX constraints such as power limits and MCS compatibility, with a subgroup-creation heuristic reducing computational complexity. The optimization maximizes the Aggregate Data Rate, while promoting fairness through balanced channel data rate (CDR) across layers. Performance is evaluated using a fitness function that favors high-throughput, MCS-aligned allocations. Simulations show notable improvements, with ADR ranging from 350 Mbps to 1575 Mbps and CDR up to 31.5 Mbps, achieving fairer resource distribution. Although PSO effectively tackles multicast scheduling and fairness, results are limited to simulations. Practical issues such as mobility, interference, and PSO parameter tuning may affect real-world performance. The approach shows strong potential but requires further validation and broader fairness analysis.

ACO is a probabilistic metaheuristic inspired by the foraging behavior of real ants. The study in [102] examines resource allocation in full-duplex (FD) multi-cell wireless networks using an ACO method, showing it outperforms greedy search approaches as network size and interference grow. The model aims to maximize the system sum rate by scheduling user pairs and selecting transmission modes either FD or half-duplex to reduce inter- and intra-cell interference while exploiting FD capacity gains. The model uses several inputs, including network configuration parameters like the number of base stations, users, and resource blocks. It also incorporates user location and path loss data, as well as power and noise parameters such as bandwidth and self-interference factors. Representing user pairs as graph nodes with interference-based edges, the model outputs an optimized resource allocation strategy that determines user scheduling, transmission mode selection, and an optimal allocation path identified through ACO. By modeling artificial ants that explore and reinforce efficient allocation routes via virtual pheromones, ACO enhances throughput, spectral efficiency, and scalability in managing FD interference. However, while offering strong adaptability and global optimization for 5G and 6G systems, ACO remains computationally intensive, parameter-sensitive, and slower to converge, limiting its practicality for real-time deployment.

Bio-inspired algorithms are also applied to content placement optimization across distributed edge caches, where the decision space is the combinatorial assignment of content items to cache locations subject to storage capacity constraints. This problem is NP-hard due to the exponential number of possible content-to-cache mappings, making it a natural fit for metaheuristic search. A hybrid PSO-GA approach [103] addresses joint edge caching and task scheduling in MEC environments, formulated as an integer non-linear programming problem accounting for device mobility and cooperative caching. In the PSO component, each particle represents a content placement and task assignment solution. Particles encode task migration decisions while cache states update dynamically using LRU based on assigned tasks. The GA component uses crossover and mutation operators to maintain diversity and escape local optima. The fitness function maximizes task completion rate, with delay calculations (transmission, migration, execution) determining deadline feasibility. Inputs include network topology, content catalogue, user mobility patterns, storage capacities, and task requirements. The output is an optimized caching and scheduling configuration. Results show the hybrid PSO-GA outperforms standalone PSO and greedy baselines across task completion rate, cache hit rate, and load balancing. However, the method is computationally intensive with convergence times potentially prohibitive for real-time decisions in dynamic environments. Performance also depends heavily on parameter tuning (population size, mutation rate, inertia weight).

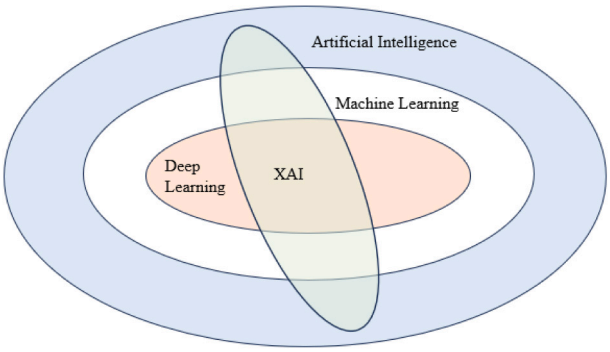


Fig. 6. Areas of AI and their mutual relationship in a Venn diagram. *Source:* Adapted from [113].

The AI-based resource allocation techniques summarized in Table 3 demonstrate significant potential for application within the highly complex landscape of future wireless communication networks. However, they require large amounts of data to provide optimal performance. Acquiring the large data sets could be either too expensive or not practical. Owing to the highly dynamic characteristics of the environment, it is challenging to acquire up-to-date channel information within the channel coherence time. Hence, according to [104], optimal resource management requires a combination of model-based and AI-based solutions rather than relying solely on one approach.

3.3. A comparative analysis of resource allocation methods

To justify the integration of AI into 6G architectures, it is essential to benchmark its performance against established model-based baselines. As detailed in Section 3.1, traditional methods such as mathematical optimization and game theory provide theoretical optimality but face severe scalability limits. This section analyzes the trade-offs between these traditional benchmarks and the AI-driven solutions surveyed in Section 3.2, focusing on three critical axes: computational complexity, adaptability to non-linear dynamics, and data dependency.

Traditional rule-based systems depend on static, human-defined rules to govern allocation decisions. While this approach ensures predictability and interpretability, it fails when network conditions shift rapidly or unpredictably. As seen in study [114], rule sets quickly lose relevance outside their original design context, making frequent manual reconfiguration necessary. This rigidity contrasts sharply with AI-driven models, which continuously learn and adjust to new traffic patterns, mobility events, and interference conditions in real time [115, 116].

Heuristic and rule-based methods struggle with the exponentially growing state and action spaces characteristic of dense 6G networks. Their combinatorial formulations often require handcrafted constraints and optimization heuristics that fail to generalize under diverse multi-cell, multi-user scenarios [117]. As computational and coordination demands rise, such methods either simplify the problem sacrificing optimality or become computationally infeasible. By contrast, AI-based mechanisms, particularly DL and RL, scale naturally through distributed training and function approximation, maintaining robust performance in large, decentralized environments [118].

Table 3
AI techniques for resource allocation.

AI category	AI technique	Application	Objective	Input/State	Output/Action	Challenges
SL	ANN	Ultra-reliable edge computing [72]	Minimize total blocking probability; Maintain QoS by keeping preemption and blocking probabilities within limits; Balance spectrum and storage resources for efficient inter-edge-node transfers.	Percentage of LC tasks; Spectrum resource; Storage; Traffic rates	Blocking probability of LC tasks; Preemption probability of LC tasks	Require extensive training data; ensure model robustness; manage ANN computation overhead; integrate with existing systems
	GNN	D2D networks [79]	Maximize weighted sum rate via power optimization; Reduced latency; Scalability; Efficiency	Directed graph where nodes represent D2D pairs and edges represent interference links; Node features; Edge features; Environmental data	Optimized transmission power for each transmitter	Complex model; ensure robustness under varying conditions; seamless integration with existing systems.
	DNN	Adaptive resource allocation responding to dynamic user demands [80]	Maximizes sum utility across active users; Ensures user utility demands are met on average; Adapts to changing constraints without retraining; improving speed and efficiency.	Channel data; Interference; Queues; History; Local channel and neighbor data rates	Transmit power levels; frequency/sub-channel assignments; timeslots; and beamformer angles.	Assumptions on demand and channel dynamics may restrict applicability; iterative process may introduce latency
	LSTM	Proactive edge caching via content popularity prediction [81]	Maximize cache hit ratio; minimize content retrieval latency under finite storage capacity	Historical content request logs; temporal features; base station load indicators	Predicted popularity ranking over content catalogue; cache placement decisions	Cold-start for new content; data staleness; vulnerability to sudden popularity shifts from viral events
	FL + Neural Network	Privacy-preserving popularity prediction for edge caching [82]	Predict file popularity for proactive caching without disclosing user request data	Local user request histories; weighted user preference vectors	Aggregated popularity estimates at the base station; cache placement policy	Communication overhead from federated rounds; model heterogeneity across users; non-IID data distributions
UL	K-means	UAV assisted networks [84]	Minimize distance between UAVs and ground users to boost channel gain; maximize data rate; reduce system delay	User coordinates; Number of clusters; Initial centers	UAV coordinates; User association; Cluster size; User pairing; Complexity	Sensitive to initial settings; assumes spherical clusters not always matching real user patterns; may overlook user mobility and dynamic channels
	K-means	D2D networks [85]	Maximize total system throughput; Minimize interference; Ensure QoS; Enhance spectrum efficiency and fairness	User info; Channel data; Constraints; Thresholds	D2D clustering; Resource allocation; Power distribution; Performance metrics	Designed for single-cell scenarios; less effective in large or mobile networks; Limited real-time adaptability for dynamic environments
	K-means	Local content popularity estimation via base station clustering [86]	Improve cache hit probability by deriving local rather than global popularity distributions	Historical request vectors per SBS; content metadata; SBS location data; cosine similarity scores	SBS clusters with similar demand patterns; per-cluster local popularity distribution; cluster-specific cache placement policy	Static clusters do not adapt to temporal popularity shifts; sensitivity to initial centroids; spherical cluster assumption may not match real demand topology
RL	Q-learning	Radio resource allocation and network slicing [96]	Maximize user satisfaction through efficient resource management; Determine optimal resource allocation ratio per network slice and BS; Select scalable numerology for diverse service needs; Meet SLAs	Resource ratio; Numerology	Agent decisions to maximize user satisfaction; Adjust resource ratio within 1–100; Adjust numerology within 0–4; Learns optimal strategy through interaction with environment	Scalability issues from large state–action spaces and long training times; LTE-WLAN integration challenges may hinder real-world deployment
	DRL	UAV assisted MEC networks [91]	Minimize weighted sum of system latency and energy use while ensuring QoS	UAV Movement Agent: UAV 3D positions, MU positions, and power allocation matrix (offloading/user association); MU Association Agent: Channel gain between MUs and MEC servers (UAVs, base station).	UAV Movement Agent: Continuous-includes movement distance and flying direction (yaw angle); MU Association Agent: Discrete-determines which UAV or base station each MU connects to for offloading.	Depends on accurate simulations; real-world issues like communication errors, imperfect data, and high training computation demands.

(*continued on next page*)

Table 3 (*continued*).

AI category	AI technique	Application	Objective	Input/State	Output/Action	Challenges
	Policy Gradient methods	UAV assisted networks [87]	Spectral efficiency; Fairness; Minimize interference	Observable environment indicators; Dynamic conditions; Channel traits; CSI	Generate subband assignment policies; Builds allocation matrix mapping users to frequency subbands for efficient resource distribution.	Scalability and real-world adaptability in diverse UAV scenarios should be further examined
	MARL	MEC networks [97]	Maximize long-term energy efficiency (transmission rate/total power)	Local observations from previous timestep: Statistical CSI (channel gain), Previous resource allocation (subcarriers, power), Previous transmission rate	Subcarrier assignment; Power control	Training complexity, coordination overhead, and potential gaps between simulation results and real-world conditions
	LSTM-DDPG	Joint computation offloading and distributed edge caching [98]	Minimize long-term average system overhead; maximize cache hit rate	Current cache contents per edge server; task/content request patterns; server computational load; backhaul link status	Offloading decisions (local/edge/cloud); cache placement and eviction decisions per edge node	High training cost for joint action space; distributed cache coherence; sensitivity to reward weight tuning between latency and backhaul objectives
	MARL with graph attention	Cooperative edge caching in vehicular networks [99]	Minimize total content access latency across all RSUs via cooperative caching	Local cache state per RSU; recent content request history; traffic flow characteristics; inter-RSU graph topology	Discrete cache/evict decisions per RSU over content catalogue, coordinated via CTDE paradigm	Scalability of graph attention to large networks; computational cost grows with number of agents; high user mobility in vehicular scenarios
Bio-inspired algorithms	GA	Heterogeneous cloud environments [100]	Maximize resource utilization (CPU, memory); Minimize network communication overhead; Improve system reliability and stability; Balance load distribution across nodes; Reduce execution and retrieval time through efficient scheduling	System entities; Relational matrices; Performance metrics; File metadata;GA parameters: Population size, mutation probability, termination iterations	Pareto front; Microservice placement; Workload distribution; Server categorization	Success hinges on accurate workload modeling and GA parameter tuning
	PSO	Mobile WiMAX networks [101]	Maximizes cell Aggregate Data Rate as sum of achievable rates over resources, balancing CDRs via throughput- and MCS-focused fitness.	WiMAX cell configuration (7 layers with MCS per user distance); user distribution across layers; number of multicast subscribers per subgroup; available OFDMA resources;PSO parameters (population size, iterations)	Optimal mapping of subcarriers/OFDM symbols to multicast subgroups, maximizing ADR and balancing CDR	Simulation-only; real-world mobility/interference untested
	ACO	IoT networks [102]	Maximize system sum rate across the multi-cell network	Network config; Channel data; Power/noise; Self-interference; Graph representation	User scheduling; Transmission mode; Optimal path and rate	Computationally intensive; parameter-sensitive; slow to converge
	Hybrid PSO-GA	Joint edge caching and task scheduling in MEC with mobility [103]	Minimize total service delay under storage capacity constraints	Network topology; content catalogue with item sizes; user mobility patterns; edge node storage capacities; task computational requirements	Optimized content-to-cache assignment matrix; task-to-server scheduling; joint caching-scheduling configuration	Computationally intensive convergence; parameter sensitivity (population size, mutation rate, inertia weight); may be prohibitive for real-time decisions

Traditional resource allocation relies on accurately specified system models and expert defined criteria, assuming predictable relationships among network variables. In real-world deployment, however, such relationships are nonlinear, time-varying, and partially observable. The inability to model uncertainty and emergent patterns limits the generalization capacity of traditional frameworks [119]. AI-driven approaches eliminate this dependence by directly learning from data and feedback, enabling decisions grounded in experiential optimization rather than predefined equations.

Conventional optimization techniques focus on instantaneous or short-term resource efficiency, neglecting cumulative or sequential effects central to 6G performance goals like energy fairness or latency stability. RL addresses this through trial-and-error exploration and reward-driven learning, enabling robust adaptation under uncertainty and across time horizons [116]. Such temporal awareness marks a critical departure from traditional short-sighted control schemes.

The comparative analysis in Tables 4 and 5 reveals a fundamental trade-off among adaptability, computational complexity, and solution quality. Table 5 spans methods from fast, explainable rule-based systems suited to static settings to highly adaptive DRL models for dynamic environments. This adaptability, however, incurs higher data demands, training overhead, and reduced explainability. Table 4 quantifies these trade-offs, showing that while methods like tabular Q-learning guarantee optimality, they scale poorly due to the "curse of dimensionality". In contrast, advanced models such as DQN approximate optimal policies in complex settings but require extensive interaction for training. Metaheuristics like GA and PSO offer a compromise, providing near-optimal solutions without training data but at the cost of a slower, iterative search process. The qualitative analysis in Table 5 further exposes a key bottleneck in current AI methods. Although DL and RL models are more adaptable than model-based approaches, they remain limited by "siloed intelligence", needing task-specific retraining and

Table 4
Quantitative comparison of AI models for resource allocation.

AI technique category	Specific model	Computational complexity (Inference)	Convergence time/Training overhead	Resource utilization (Solution quality)
SL	ANN/RNN [105]	Low (Forward pass: $\approx O(\sum_{l=1}^{L} n_{l-1} n_l)$, where: L is number of layers and n_l is number of neurons in each layer)	High (Requires large, labeled datasets; offline training)	Sub-optimal to Near-optimal (Performance is capped by the quality of the training data)
	Random Forest	Low to Medium (Depends on the number and depth of trees)	Medium	Sub-optimal (Prone to learning from biased or sub-optimal datasets)
UL	K-means/Principal Component Analysis (PCA) [106,107]	Low (For PCA: $O(N \cdot D)$, where: N is the number of data samples/rows and D is the number of features/columns, For K-means: $O(N \cdot k \cdot I)$, where: N is the number of data samples, k is the number of clusters specified, and I is the number of iterations until convergence)	Low to Medium	Not directly applicable (Used for pre-processing, e.g., user clustering or feature extraction, not allocation)
RL	Q-Learning [108]	Very low (Table lookup: $O(A)$ where: A is the number of available actions)	Very high (Infeasible for large state–action spaces; "curse of dimensionality")	Optimal (Guaranteed to converge to optimal policy for finite Markov Decision Processes)
	DQN/Double DQN/Dueling DQN [109]	Low (Same as ANN forward pass)	High (Requires extensive environment interaction; sample-inefficient and can be unstable)	Near-optimal (Can approximate optimal policy in complex environments but lacks theoretical guarantees)
	Upper Confidence Bound (UCB)-MAB [110]	Very fast (Simple update rule)	Very fast (Sample-efficient; suitable for online adaptation)	Sub-optimal (Effective for simpler state–action problems but over-simplifies complex system dynamics)
Metaheuristics	GA [111]	High ($O(I \cdot P \cdot C)$, where: P is population size, C is cost of fitness evaluation)	N/A (Iterative search, not training; convergence time can be long and variable)	Near-optimal (Effective global search heuristic but no optimality guarantee)
	PSO [112]	Medium ($O(I \cdot P \cdot D)$, where: I is the Number of iterations, P is the Population size, and D is the Dimension of each particle)	N/A (Convergence can be faster than GA but may fall into local optima)	Near-optimal
	ACO [58]	High (Depends on graph size and number of ants)	N/A (Iterative search with significant computational cost per iteration)	Near-optimal

Table 5
Qualitative comparison of resource allocation methods.

Method	Strengths	Limitations (Critical for 6G)	Representative use cases
Rule-based	Fast decision-making; fully interpretable	Inflexible to dynamic conditions; incapable of real-time adaptation; scalability collapse beyond small networks	Legacy systems; static or low-mobility environments [114]
Heuristic optimization	Practical in constrained, well-modeled scenarios	Cannot handle exponential state growth or nonstationary conditions; fails under large-scale 6G workloads	Localized scheduling; restricted topology optimization [117]
ML	Learns adaptive policies from data; improves over time	Requires extensive labeled data and retraining; limited generalization in unseen conditions; moderate reactivity to real-time changes	Predictive network analytics; cloud task allocation [119]
DL	Models high-dimensional, nonlinear interactions effectively	Computationally prohibitive for edge deployments; dependent on massive data; lacks rapid adaptability and transparency	Spectrum optimization; large-scale IoT networks [118]
RL	Excels in sequential and uncertain environments; autonomous optimization	High training cost; instability in sparse or complex environments; low explainability	Network slicing; real-time orchestration; hybrid cloud control [116]

labeled data for each new environment. As 6G networks expand in scale and complexity, this lack of generalization becomes increasingly restrictive. This limitation sets the stage for the paradigm shift outlined in the next section the move toward BAIM (Section 4), which use large-scale pre-training to achieve zero-shot generalization and holistic network orchestration.

4. Big AI models in 6G resource allocation

While the RL and DL agents analyzed in Section 3 demonstrate powerful adaptability in specific environments, they suffer from a fundamental limitation: poor generalization. As noted in the comparative analysis (Table 5), a DRL agent trained to optimize spectrum

in a dense urban RAN often fails when deployed in a rural SAGIN environment without extensive, computationally expensive retraining. Section 4 examines BAIM (including large language models (LLMs) and wireless big AI models (wBAIMs)) not merely as a new technology, but as the necessary architectural evolution to bridge this "generalization gap". By leveraging massive pre-training, these models aim to move 6G management from isolated, task-specific agents to holistic, adaptable intelligence [120].

4.1. Key concepts of big AI models

BAIM in 6G networks encompass several types with distinct roles enhancing future wireless communications. One category includes LLMs, such as ChatGPT and generalized pretrained transformers. These models are known for their powerful language understanding, contextual reasoning, and multi-modal processing capabilities. These models, originally developed for natural language tasks, are specialized and adapted as mobile network-specific models, often called wBAIMs. Additionally, generative AI (GenAI) models designed specifically for edge–cloud operations play a vital role by enabling local, lightweight, and flexible AI deployment close to the user or edge devices [120,121] a capability critical for efficient edge inference in wireless networks [122].

Moving beyond passive inference, recent frameworks propose "agentic AI" networking, where Generative Foundation Models function as autonomous agents. In this "model-as-agent" approach, large models are transformed into active entities that operate through continuous action loops [22]. These agents perceive network states and reason dynamically to achieve complex operational goals. Furthermore, they utilize network APIs or external solvers as tools to execute tasks without human intervention. Building on this paradigm, the AGILE-6G framework explicitly applies agentic AI to the autonomous, closed-loop management of 6G network and application services [123]. Unlike traditional orchestration, this empowers AI agents to autonomously govern the entire service lifecycle. Furthermore, bringing this autonomy closer to the user, [124] propose an "edge agentic AI framework" specifically for O-RAN. By deploying autonomous agents directly at the network edge, this approach enables real-time network optimization that addresses the strict latency constraints of 6G. Ultimately, these developments pave the way for edge general intelligence (EGI), where the "agentification" of network resources allows 6G systems to autonomously reason and adapt to open-world challenges that go far beyond pre-defined optimization tasks [125].

In the context of 6G, these BAIM perform critical functions including real-time radio resource allocation in dynamic and vehicular environments [126,127], user-centric power scheduling [128], and collaborative decision-making. For instance, [127] demonstrate that LLMs can be utilized to adaptively optimize resource allocation in dynamic wireless environments, effectively replacing traditional heuristics with LLM-driven reasoning for complex vehicular scenarios. Furthermore, they extend automation beyond terrestrial boundaries, facilitating holistic management across integrated satellite-aerial-terrestrial networks [129]. This integrated intelligence leads to smarter, faster, and more adaptive network management that can meet the extremely stringent latency, reliability, and performance demands of 6G-enabled applications [130].

Regarding deployment, the models are implemented in diverse environments to balance latency and computational resource constraints. These models can be deployed at the network edge to ensure low-latency, context-aware decision-making. Alternatively, they can be housed in centralized cloud platforms for powerful large-scale computation. Furthermore, hybrid architectures combining both approaches can be used to optimize performance and scalability. This deployment flexibility makes BAIM indispensable for delivering intelligent, efficient, and scalable resource allocation strategies that will be the hallmark of 6G wireless networks [121,131].

4.2. Limitations addressed by big AI models

DRL methods, as discussed in Section 3.2.3, often require extensive interaction data to converge toward optimal policies. BAIM address this limitation through few-shot and zero-shot learning. Unlike conventional agents that effectively "restart" when network conditions change significantly, foundational models can infer optimal resource allocation strategies for previously unseen scenarios (e.g., a sudden traffic surge in a network slice). They achieve this through context-aware reasoning rather than iterative trial-and-error [120].

Another major advantage lies in their autonomous adaptation. BAIM can ingest and synthesize multisource data such as real-time network traffic, user behavior, environmental conditions, and hardware states. By modeling this complex information holistically, they can automatically adjust resource allocation strategies to maintain performance objectives amid fluctuating network conditions. For example, they can dynamically manage spectrum, computing resources, and traffic routing with minimal human intervention. This self-evolving capability is particularly vital for 6G networks, which must accommodate extreme device densities and highly heterogeneous applications while maintaining ultra-low latency [132].

Beyond adaptation, these models can serve as high-level orchestrators for the specialized agents surveyed in Section 3. A hierarchical relationship emerges: while the DRL agent optimizes millisecond-level beamforming, a central LLM interprets broad SLAs to continuously refine the agent's reward function. In doing so, BAIM provide the context and intent, while traditional AI agents execute the fine-grained optimization, creating a synergistic framework [27].

4.3. Challenges and research directions of using big AI models

BAIM integration into 6G networks faces several key challenges. First, ensuring smooth operation across distributed edge and cloud systems while enabling end-to-end AI service orchestration requires new flexible network architectures. These architectures must support native AI capabilities, real-time collaboration, and intelligence sharing among diverse network nodes [132].

Second, latency and computing limits are critical, especially for real-time tasks like augmented reality and autonomous driving. Large AI models must be optimized with techniques such as model compression and hardware acceleration to perform fast, low-latency inference near the network edge where computing resources are limited [121].

Third, scalability and monetization remain important. Efficiently scaling AI across vast 6G deployments demands innovations in distributed learning and resource sharing. Cost-effective model deployment and new AI-driven business models are needed for operators to generate sustainable revenue [120].

Finally, further research on explainability and computational efficiency is advancing to make these models practical. XAI helps operators understand and trust model decisions, while optimization methods reduce their computational footprint for real-time edge deployment [133].

In summary, integrating BAIM in 6G requires overcoming several hurdles, including architectural complexity, latency constraints, scalability challenges, and operational transparency. Addressing these issues is essential to enable intelligent, responsive, and cost-effective next-generation networks (Fig. 7).

While BAIM (Section 4.1) successfully address the generalization gap identified in Section 3, our analysis highlights a new, critical vulnerability: the "Accuracy-Interpretability Paradox". As models grow in parameter size (e.g., large wBAIMs) to capture complex 6G correlations, their decision boundaries become increasingly opaque. In mission-critical 6G use cases such as ultra-reliable slicing for autonomous driving high accuracy is insufficient; operators require verifiable reasoning to certify safety and compliance. Consequently, the deployment of BAIM effectively trades operational transparency for performance. This paradox necessitates an architectural "trust layer", leading directly to the requirement for XAI frameworks analyzed in Section 5.

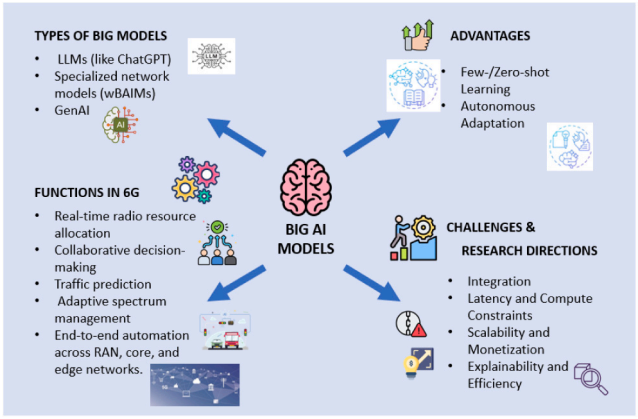


Fig. 7. Big AI models for 6G resource allocation: Concepts, Advantages, and Challenges.

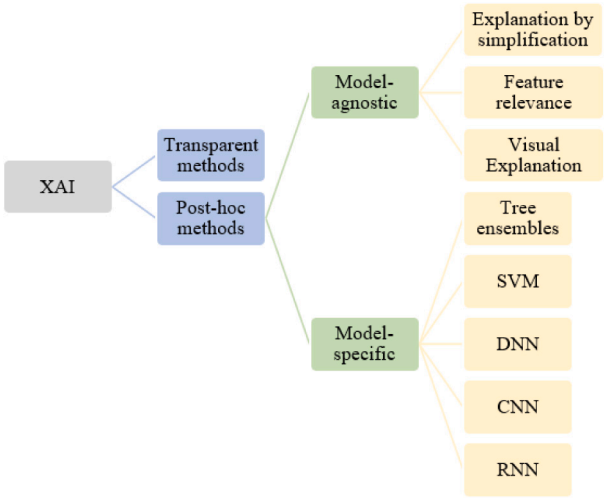


Fig. 8. Classification of XAI models [136].

5. Explainable artificial intelligence in 6G resource allocation

While the DRL agents surveyed in Section 3 and the foundational BAIM discussed in Section 4 provide the necessary computational intelligence to handle MEC complexity, they introduce a critical trade-off: opacity. These sophisticated architectures effectively function as "black boxes". This opacity makes it challenging for network operators to interpret the rationale behind specific resource allocation decisions. Consequently, it becomes difficult to troubleshoot anomalous behaviors or fully trust the network's autonomous decision-making processes. [134]. Therefore, XAI is not merely an optional feature but a mandatory architectural bridge. It translates the high-dimensional internal states of the models from Sections 3 and 4 into interpretable insights. This ensures that enhanced performance does not come at the cost of operational control.

XAI is an emerging and transversal requirement across all AI domains (Fig. 6 depicts the different areas of AI and their mutual relationship in a Venn diagram). Since AI algorithms come with the cost of lower explainability due to their obscure or complex nature, understandable interfaces are needed for the black box models [135, 136]. Here explainability refers to a model with a decent proxy to an actual algorithm that is understandable to a human being [136]. XAI is also used to improve model performance, detecting biases, and informing end users or outcomes [136]. Furthermore, the necessity for explainability in AI models can be identified [137]:

- To guarantee the reliability of the model and instill confidence that it will consistently operate as intended, regardless of the input provided;
- To find the causal relationship between input data variables;
- To transfer models between one another and understand the model constraints;
- To provide information on the problem solved and the factors considered to generate the result; and
- To highlight biases in the algorithm and ensure fairness of the decision taken.

Stakeholders including model developers, end users, and or regulatory entities can benefit from XAI. The goal of explainability can be determined by the target audience [136]. According to the study in [137], XAI can be broadly classified into two categories: Transparent methods and Post-hoc methods (Fig. 8). In this context, transparent methods pertain to ML techniques such as logistic regression and K-nearest neighbors, as well as rule-based learning approaches including generalized additive models and Bayesian models. However, the transparency property can be hindered if the depth or the amount of data is excessively large. The post-hoc methods are further classified into model-agnostic and model-specific depending on the type of algorithm used. The model agnostic methods analyze the inputs and outputs of the black box model. The explanation provided by these models will be simplification, feature relevance, or visual explanations [137]. Explanation by simplification generates simpler models for the complex black box models. Local Interpretable Model-Agnostic Explanations (LIME) [138] and Dialogue Relation Extraction with EXplanations (D-REX) [139] exemplify this type of methodology. The feature relevance approach assesses the influence of input features on the model's output decisions. SHAPley additive explanations (SHAP) is an example of this category which uses game theory concepts to calculate scores for the contribution of the features [140]. As the term implies, visual explanation techniques elucidate the internal mechanisms of a model by offering interpretability through graphical or visual representations. The Partial Dependence Plots (PDPs) can be considered under this category [141]. Conversely, model-specific methods leverage the inherent characteristics of particular ML or AI models to generate explanations. As a result, these techniques are tailored to, and can be categorized according to, the underlying AI model [136,138–141].

5.1. XAI in multi-access edge computing

MEC, brings cloud computing capabilities (storage, processing, analytics) closer to end-users by deploying resources at the network edge rather than in centralized data centers. This reduces latency, saves bandwidth, and enables real-time decision-making, which is vital for time-sensitive use cases like autonomous vehicles and industrial IoT [134,142]. MEC is not just an add-on but a foundational technology for 5G and future 6G networks. It is the key enabler for the most demanding 5G or 6G service categories, including URLLC for applications like autonomous driving and industrial robotics, eMBB for services like cloud gaming and Artificial Reality (AR) or Virtual Reality (VR), and mMTC for large-scale IoT deployments [143].

5.1.1. XAI-based resource management using network slicing in MEC

In 5G and beyond, network slicing enables operators to create multiple virtualized, end-to-end networks over a shared physical infrastructure. Each slice is optimized to meet distinct QoS requirements dictated by specific applications, such as URLLC or high throughput for eMBB. AI, and in particular DRL, provides a robust framework for automating the inherently complex and dynamic processes of resource allocation and computation offloading essential for the efficient management of these network slices.

Table 6
Summary of XAI approaches in MEC.

Domain	Specific task	AI model	XAI method	Contribution
Network slicing	Vehicular network slicing [144]	DRL	SHAP and attention mechanism	Enhanced QoS fulfillment for URLLC and eMBB applications; provided real-time insights into allocation decisions
	Multi-agent network slicing [145]	Multi-Agent DRL	PVDN	Explained reward trade-offs between slices; significantly improved throughput and reduced latency
	RAN-Edge resource provisioning [146]	FL	Integrated gradients	Enabled trustworthy client selection in FL, improving model convergence and resource provisioning
Computation offloading	General/IoT [133]	XGBoost	SHAP	Improved energy and delay performance
	V2X [147]	Multi-Agent DRL	SHAP	Sum-rate maximization and adherence to reliability constraints across both V2N and V2V links
Model compression & rule extraction	IoT [148]	Deep Belief Networks (DBNs)	Rule extraction from pruned DBNs	Delivers higher accuracy than other rule extraction methods, bridging data/hardware demands and edge/IoT limits with compact, explainable generative models
	Image classification [149]	EfficientNet B2 and EfficientNet V2	Gradient-based feature saliency	Introduces T-SHIELD and X-SHIELD, integrating XAI-guided feature hiding into loss for better accuracy, convergence, and explanation quality.
Hardware acceleration	Accelerating computationally intensive XAI operations [150]	Versatile	Approximate computing applied to XAI algorithms	Introduces XAI edge framework, achieving 2× energy efficiency improvement over accurate XAI hardware methods on TPU-based edge devices while maintaining comparable accuracy.
Model efficiency and interpretability	Neuromorphic computing [151]	Spiking Neural Network (SNN)	SHAP, LIME, and eli5	XAI-based SNN framework surpassing DNNs in accuracy, efficiency, and interpretability for trustworthy healthcare deployment.
	Industrial process modeling [152]	Linear Regression, 3-layer Neural Network (ReLU, PyTorch), and XGBoost regressor	SHAP (model-specific variants), LIME, Gradient baseline, Smooth Gradients (SG), Averaged Local Effects (ALE-kNN);	Proposes ground-truth sensitivity metrics to assess XAI robustness, showing effect-based methods outperform additive ones under noise, linked to model performance.

Despite their effectiveness, DRL-based agents often function as opaque "black boxes", making it challenging for network operators to interpret the rationale behind specific resource allocation decisions, troubleshoot anomalous behaviors, or fully trust the autonomous decision-making processes [153]. To address this opacity, the field of Explainable Reinforcement Learning (XRL) is being applied to MEC and RAN environments. For instance, [154] propose an "Explanation-guided Deep Reinforcement Learning" framework specifically for trustworthy 6G RAN slicing. By integrating explanation modules directly into the learning loop, this approach ensures that the agent's slicing decisions are not only optimal but also verifiable, fostering the necessary trust for automated network management.

Complementing these architectural frameworks, feature-based XAI methods, such as Shapley values combined with attention mechanisms, are being developed for vehicular network slicing. These systems provide real-time insights into which factors (e.g., vehicle density, channel quality) most influenced a resource allocation decision. Studies have shown that this transparency not only builds trust but can also be used to refine the agent's policy, leading to improved QoS satisfaction for both URLLC and eMBB services [144].

In multi-agent systems where different agents manage different slices, XAI-driven approaches such as Prioritized Value-Decomposition Networks (PVDN) are used. PVDN explains the system's behavior by decomposing the global reward into the individual contributions of each agent. This allows an operator to understand the trade-offs being made, for example, how prioritizing latency for the URLLC slice impacts the throughput of the eMBB slice. This explainability has been shown to enable better optimization, with PVDN significantly outperforming non-explainable VDN and independent learning methods in both throughput and latency [145].

When resource management is handled via FL at the RAN-Edge to preserve data privacy, XAI plays a crucial role in improving the training process itself. By applying techniques like integrated gradients, the central aggregator can assess the importance of features from each client's local data. This information is then used to guide client selection, prioritizing those that provide the most valuable data contributions. This XAI-driven selection process enhances the trustworthiness and accelerates the convergence of the global model for slice-level resource provisioning [146].

Complementing these control mechanisms, XAI is also proving vital in the design of the predictive models themselves. For cloud-native 6G networks, [155] introduce an "XAI-driven model design" approach for resource utilization forecasting. This framework utilizes interpretability techniques during the model development phase to identify critical features. As a result, it creates forecasting models that are accurate, lightweight, and transparent. These qualities are essential for the dynamic scaling of network slices.

5.1.2. XAI-based computation offloading in MEC

Offloading in MEC allows resource-constrained devices such as IoT sensors, mobile phones, and UAVs to shift computationally intensive

tasks to nearby edge servers, optimizing performance by managing trade-offs among latency, energy consumption, and overall system efficiency. These offloading decisions must continuously adapt in real time to fluctuating factors, including network latency, user mobility, channel conditions, and varying edge server workloads [156].

In [133], the authors propose a novel framework for optimizing computation offloading in MEC environments by integrating XAI techniques. Specifically, they model the offloading decision as a multi-label classification problem and utilize the eXtreme Gradient Boosting (XGBoost) algorithm in conjunction with SHAP to enhance both predictive accuracy and model interpretability. The study demonstrates that SHAP can be effectively leveraged for feature selection, resulting in improved classification performance and reduced system complexity. Simulation results show that the proposed XAI-assisted approach significantly lowers system costs and energy consumption when compared to conventional offloading strategies.

In [147], the authors propose a framework that leverages XAI, specifically SHAP, to enhance the interpretability and reduce the complexity of DRL models for resource allocation in cellular V2X communication networks. The study addresses the inherent "black-box" nature of DRL models, which poses a significant barrier to their practical deployment in safety-critical applications due to the lack of transparency in decision-making processes. To tackle this challenge, the authors introduce a two-stage, model-agnostic methodology. In the first stage, SHAP scores are employed to identify and rank the importance of input state features. In the second stage, these rankings are used to prune less relevant features, thereby simplifying the model and reducing computational overhead while preserving high performance. The proposed approach effectively balances model interpretability and performance, leading to reductions in training time and complexity. Experimental results demonstrate the framework's ability to maintain strong network performance in V2X communication scenarios, addressing key objectives such as sum-rate maximization and adherence to reliability constraints across both Vehicle-to-Network (V2N) and Vehicle-to-Vehicle (V2V) links.

Most existing research focuses on explaining binary offloading decisions (e.g., local vs. edge). However, as 6G applications become increasingly cloud-native, the challenge shifts to managing the decomposed components of these applications. Addressing this, [157] propose an XAI-driven framework specifically for 'Resource-Efficient Microservice Management' in cloud-native networks. By applying interpretable ML to the orchestration of microservices, this approach allows operators to understand and optimize the trade-offs between resource consumption and service latency at a granular level, ensuring that the flexibility of microservices does not come at the cost of opacity.

A comprehensive summary of these and other XAI approaches in MEC is presented in Table 6. Collectively, these works highlight that integrating XAI into MEC-based offloading is crucial not just for performance, but for the safety, reliability, and user acceptance of edge computing systems. However, despite notable progress in AI-based offloading strategies, the explicit incorporation of explanatory frameworks remains relatively rare, marking a significant gap in the current literature.

Furthermore,integrating XAI into the 6G resource management loop closes the trust gap, but our review identifies a lingering challenge: the "Cost of Explanation". Techniques like SHAP (Section 5.1.2) introduce significant computational overhead, which can be counter-productive for latency-sensitive MEC applications (e.g., sub-millisecond tactile internet). Therefore, a key insight from this survey is that 6G management cannot rely on "full explainability" for every micro-decision. Future architectures must likely adopt a selective XAI approach providing real-time lightweight explanations for frequent actions while reserving deep, post-hoc analysis for critical failure events. This balance between trust and latency remains the definitive open challenge for the 6G-AI ecosystem.

5.2. XAI-complementary digital twins for resource allocation validation

XAI enhances interpretability for trustworthy autonomous operation. However, deploying AI-driven resource allocation agents, particularly the BAIM discussed in Section 4.1, poses a key challenge. These models require extensive interaction with complex network environments during training. Direct exploration on live infrastructure, however, risks service degradation and SLA violations [158,159]. DTs have emerged as core enablers of AI-native 6G networks, offering high-fidelity, real-time virtual replicas of network infrastructure spanning the 3C resource continuum [160,161]. A Network Digital Twin (NDT) provides a synchronized digital replica of the mobile network, supporting real-time monitoring, prediction, simulation, and optimization without affecting operations [161,162]. In AI-driven 6G resource allocation, DTs serve three complementary roles. First, they offer a safe virtual environment for training and calibrating AI agents before deployment [158,159]. Second, they enable what-if analysis to assess alternative resource management strategies across heterogeneous domains [161,163]. Third, they improve explainability through counterfactual simulation to complement post-hoc XAI methods like SHAP [164]. This final role mitigates the computational overhead of real-time explanations in latency-sensitive MEC scenarios. Table 7 summarizes these roles alongside the relevant AI models and 3C resource domains.

5.2.1. Safe training and validation

As outlined in Section 3.2.3, DRL agents require extensive network interaction to converge on optimal resource allocation policies. Deploying these agents directly on live 6G infrastructure poses significant risk, as stochastic exploration during RL training may cause suboptimal actions that degrade QoS, breach SLA constraints, or destabilize services [158,159]. This issue is heightened in 6G, where the scale and heterogeneity of the 3C resource space magnify the impact of exploratory failures across spectrum, computing, and caching domains.

DTs mitigate this risk by providing high-fidelity virtual environments for training, calibration, and stress testing before live deployment. A DT replicates network dynamics using historical data and neural network models, forming a closed-loop training system in which the DRL agent interacts with a digital replica rather than the operational network [158]. Within this sandbox, operators can emulate diverse scenarios, such as IoT traffic surges, rapid SAGIN mobility, or simultaneous slice admissions, allowing agents to develop robust allocation strategies without impacting service continuity. [158] showed that this approach improves convergence and performance of DRL-based slicing policies relative to direct environment training, while [159] demonstrated DT-enhanced RL supports pre-evaluation of allocation behaviors across digital domains for safer real-world deployment. Likewise, [165] introduced a DT-accelerated online DRL framework for slice admission control that increased resource utilization by up to 26% compared with state-of-the-art baseline models.

This offline training capability is essential for zero-touch network management in 6G, ensuring AI models achieve operational readiness before deployment at the edge or core. By decoupling training from live infrastructure, DTs reduce the cost and risk of automation while accelerating the development of the specialized agents discussed in Section 3.2.

5.2.2. What-if analysis and root-cause diagnosis

Beyond safe training, DTs enable operators to prospectively evaluate AI-driven resource allocation decisions. Before implementing complex 3C resource reconfigurations recommended by BAIM (Section 4), proposed actions can be simulated within a DT to predict their effects on QoS, latency, energy efficiency, and SLA compliance [161,163]. [161] highlight that DTs allow researchers and operators to emulate configurations and identify optimal strategies when direct experimentation on live networks is too risky. This what-if simulation is especially vital for

Table 7
Summary of digital twin functions for AI-driven resource allocation.

DT function	Specific task	AI model	3C domain	Contribution
Safe training & validation	DRL pre-validation for network slicing [158]	DQN, DDQN, Dueling GAN-DDQN	Communication	Dual closed-loop architecture enabling DRL agents to train in DT before fine-tuning on live network; improved convergence speed and policy quality
	DT-accelerated admission control [165]	Modified A2C (Actor-Critic)	Communication, Computing	Behavior cloning-based DT accelerates online DRL; 26.39% resource utilization improvement over directly trained models
	Safe RL for vehicular edge computing [159]	DRL (multi-domain)	Communication, Computing	DT enables safety evaluation of offloading strategies across digital domains before real-world execution
What-if analysis & root-cause diagnosis	What-if scenario analysis for 6G wireless management [166]	Scenario-based simulation	Communication	Scenario-maker module evaluates alternative CST and TPC configurations; demonstrated scalability across varying network topologies
	AI-enabled traffic forecasting and proactive resource management [163]	LSTM	Communication, Caching	LSTM-integrated DT framework for traffic prediction and proactive allocation; outperforms baseline methods
Enhanced explainability & trust	GenAI-driven counterfactual simulation [164]	GenAI (Transformer, Diffusion)	Communication, Computing, Caching	High-fidelity DT powered by GenAI enables operators to visualize and validate AI-driven decisions across 6G slice types
	GenAI-empowered network DTs [161]	GenAI, GNN	Communication, Computing, Caching	Comprehensive NDT framework supporting what-if analysis, anomaly detection, and counterfactual reasoning for transparent network operations

URLLC services, where transient performance loss is unacceptable, and for SAGIN environments (Section 1.1), where real-world testing across satellite, aerial, and terrestrial systems is prohibitively complex.

Recent studies have formalized this capability through dedicated frameworks. For example, the what-if analysis framework in [166] includes a scenario-maker module that generates hypothetical configurations and evaluates them via a DT-integrated simulation layer. This allows comparison of alternative strategies, such as channel selection or transmission power control, before applying changes to the physical network, demonstrating scalability across diverse 6G topologies.

Complementing prospective evaluation, DTs also support retrospective root-cause diagnosis. Maintaining synchronized historical network states [161,162] enables event replay to isolate factors such as resource scarcity, suboptimal algorithms, or environmental interference. This diagnostic capacity enhances validation of opaque AI models by distinguishing between provisioning errors and faulty policies learned by DRL agents. Thus, DT-based root-cause analysis complements the post-hoc XAI techniques in Section 5.1 by providing temporal and causal insights beyond the scope of feature-attribution methods like SHAP.

5.2.3. Enhancing explainability and trust

A central finding of this survey is the "Cost of Explanation" challenge identified in Section 5.1.2: real-time XAI methods such as SHAP incur significant computational overhead, undermining their applicability in latency-sensitive MEC environments. DTs offer a complementary solution through two key capabilities: counterfactual reasoning and outcome visualization.

DTs enable counterfactual explainability by allowing operators to ask, "what would have happened under an alternative allocation?" Instead of relying solely on feature-attribution scores, operators can replay an AI agent's decision within the DT under modified conditions, e.g., testing how a different spectrum allocation would influence interference in a SAGIN scenario. This approach provides causal insight beyond post-hoc methods such as SHAP or LIME, which quantify correlations but not temporal or causal dynamics [161,164]. Crucially, DT-based counterfactuals can be executed offline, eliminating latency penalties and directly mitigating the cost-of-explanation bottleneck.

DTs also act as visualization interfaces that translate abstract explanations into interpretable operational contexts. While SHAP values quantify feature influence, their meaning may remain opaque to non-expert operators. By correlating such explanations with simulated outcomes, e.g., visualizing how a computation offloading decision affects edge utilization and latency, DTs enhance understanding of AI decision rationales [160,164]. [164] illustrate this through a GenAI–driven DT framework that simulates network behavior under diverse 6G conditions, enabling explanation validation before real execution. This fusion of reasoning and simulation closes the trust loop, supporting autonomous, zero-touch 6G resource management across the 3C continuum.

Despite their promise, integrating DTs into AI-driven 6G resource management presents several open challenges. First, synchronization fidelity is critical: DT utility depends on accurately mirroring the physical network, yet maintaining real-time, bidirectional alignment in dynamic 6G environments with high user mobility and heterogeneous SAGIN topologies remains difficult [160,161]. Any deviation between the twin and live network can result in AI policies that perform well in simulation but degrade in deployment [158,159]. Second, constructing and sustaining high-fidelity DTs across the 3C resource continuum demands significant computational resources, potentially conflicting with 6G energy-efficiency goals and motivating research into lightweight or approximate DT models [160,162]. Third, ensuring DT security and trust is essential, as adversarial threats such as data poisoning during digital mapping could compromise AI policies trained within the twin [159,162]. Addressing these challenges through privacy-preserving federated learning and energy-efficient DT architectures forms a key direction for future research.

In summary, as 6G networks increasingly rely on highly complex, autonomous AI models, establishing a robust trust and validation layer is imperative. This section has demonstrated that XAI is vital for decoding the "black-box" decisions of deep reinforcement learning and BAIM. However, the computational overhead or "cost of explanation" associated with real-time XAI methods presents a significant bottleneck for latency-sensitive 6G services. To resolve this, DTs emerge as an indispensable complementary technology. By providing

high-fidelity virtual environments, DTs enable the safe offline training of AI agents, prospective what-if scenario testing, and counterfactual explainability without impacting real-time network performance. Ultimately, the synergistic integration of XAI and DTs bridges the gap between high-performance AI autonomy and operational transparency, laying the necessary foundation for trustworthy, zero-touch 6G resource management.

6. Discussion and conclusion

The preceding sections have systematically traced the evolution of resource allocation in 6G networks. This progression began with the foundational 3C framework in Section 2. It then moved through the limitations of model-based approaches and the rise of AI-driven techniques in Section 3. Finally, it explored the emergence of BAIM and agentic AI paradigms for holistic orchestration in Section 4 and the integration of XAI and DTs as an essential trust and validation layer in Section 5. While this progression reveals a clear trajectory toward increasingly intelligent and autonomous network management, it is equally important to critically assess what the current body of research has not yet achieved. This section first consolidates the cross-cutting limitations that persist across the surveyed literature (Section 6.1), then outlines future research directions and presents concluding remarks (Section 6.2).

6.1. Current limitations of AI-based 6G resource allocation

Despite the significant progress documented in this survey, the existing literature on AI-based resource allocation for 6G exhibits several cross-cutting limitations that constrain the maturity and deployability of current solutions.

- **Scalability beyond simulation:** The vast majority of the AI techniques surveyed in Section 3, from DRL-based spectrum allocation to MARL-based MEC optimization, have been validated exclusively in simulated environments with controlled parameters (as noted in the challenge columns of Table 3). Performance in simulations with tens of users or a handful of base stations does not reliably predict behavior in production 6G networks expected to serve millions of devices across heterogeneous infrastructure. The scalability gap between controlled experimentation and real-world deployment remains largely unaddressed across all AI categories.
- **Cross-Domain Scalability and Orchestration:** The integration of heterogeneous networks (e.g., terrestrial RAN and rural SAGIN) introduces a two-fold limitation for current AI frameworks. First, from an algorithmic perspective, agents exhibit "siloed intelligence"; a model trained for urban spectrum management typically fails to generalize to a satellite environment without extensive retraining. Second, from an orchestration perspective, joint optimization across the space–air–ground continuum, where latency profiles and computational capacities differ radically, remains largely unaddressed. Current models struggle to simultaneously coordinate these highly diverse physical resources at scale.
- **Real-time operational feasibility:** A persistent tension exists between model complexity and inference latency. The deep learning and foundation models that achieve the highest accuracy often impose computational overhead incompatible with the sub-millisecond decision timescales required by URLLC and tactile internet applications. Lightweight alternatives such as the AERO framework [73] demonstrate promising directions but remain limited to specific forecasting tasks. The literature has not yet established general-purpose methods for achieving simultaneously high accuracy and real-time inference across the full 3C resource continuum.
- **Data availability and quality:** Effective AI-driven allocation depends on access to large volumes of high-quality, labeled training data, channel state information, traffic traces, mobility patterns, and workload profiles. In practice, such datasets are proprietary, costly, geographically biased, and rapidly obsolescent in dynamic 6G environments. While federated learning [82] and DTs (Section 5.2) offer partial mitigation, the cold-start problem for newly deployed networks and the challenge of data representativeness across diverse operating conditions remain open.
- **Explainability-performance trade-off:** As identified in Section 5.1.2, the "Cost of Explanation" remains unresolved. The XAI techniques that provide the deepest interpretability (e.g., SHAP-based feature attribution) also introduce the most computational overhead. The field lacks established methodologies for selective explainability, determining which decisions require real-time explanation versus post-hoc audit, and for quantifying how much accuracy can acceptably be traded for transparency in different 6G service contexts.

Collectively, these limitations indicate that while AI-based 6G resource allocation has demonstrated compelling proof-of-concept results, the transition from algorithmic innovation to operational deployment requires addressing fundamental challenges in scalability validation, cross-domain generalization, real-time feasibility, data governance, and trustworthy autonomy.

6.2. Future research directions and concluding remarks

Addressing the limitations identified in Section 6.1 requires targeted advances across several interconnected research fronts. First, distributed orchestration must replace centralized control to overcome scalability and multi-domain deployment gaps. This requires scalable multi-agent frameworks capable of managing resources across terrestrial and non-terrestrial nodes in SAGIN environments. The agentic AI paradigm discussed in Section 4.1 offers a promising foundation, but validation must extend beyond single-domain prototypes to encompass joint SAGS resource coordination at realistic scale.

Second, reconciling the tension between model complexity and real-time feasibility demands energy-efficient and latency-aware algorithms. Techniques such as model compression, knowledge distillation, and hardware-aware neural architecture search must be tailored specifically for the 3C resource continuum. This tailoring ensures that inference latency does not compromise the sub-millisecond requirements of URLLC services.

Third, addressing data scarcity and generalization gaps requires federated learning combined with digital twin-based synthetic data generation (Section 5.2). However, standardized benchmarks and open datasets for 6G resource allocation research are urgently needed to enable reproducible cross-study comparisons.

Fourth, robust security mechanisms must defend against adversarial attacks and data poisoning, ensuring AI-driven autonomy remains resilient and secure across all network architecture layers.

As the 3C framework evolves to accommodate sensing, AI-driven resource allocation for ISAC presents distinct challenges unaddressed by communication-only frameworks. Unlike conventional spectrum allocation, ISAC resource management requires simultaneous optimization of dual-functional objectives: sensing metrics (detection probability, target localization accuracy) alongside communication metrics (throughput, latency). This creates a fundamentally different reward structure for RL-based approaches, where sensing-communication trade-offs must be dynamically balanced based on application context. Additionally, cooperative sensing across distributed base stations introduces multi-agent coordination challenges analogous to, yet structurally distinct from, cooperative caching and computation offloading. Developing unified AI frameworks that jointly orchestrate the emerging "3C+S"

resource continuum, spanning communication, computing, caching, and sensing, represents a critical research frontier.

Finally, the explainability-performance trade-off demands a paradigm shift from universal to selective explainability. Future 6G architectures should adopt tiered XAI strategies that provide lightweight real-time explanations for routine allocation decisions while reserving computationally intensive post-hoc analysis for critical failure events and regulatory audits. Integrating DT-based counterfactual reasoning (Section 5.2.3) with conventional feature-attribution methods offers a promising hybrid approach.

Ultimately, the co-evolution of agentic AI, DTs, and XAI will define a 6G ecosystem that balances extreme operational autonomy with strict interpretability and trust. By mastering this balance across the 3C resource continuum, operators can pave the way for a secure, resilient, and fully intelligent IoE.

CRediT authorship contribution statement

Rasini P. Amarasooriya: Writing – original draft, Methodology, Conceptualization. **Mark A. Gregory:** Writing – review & editing. **Shuo Li:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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